CamCAN Selectivity & Attrition Analysis (for *The Cambridge Centre for Ageing and Neuroscience (Cam-CAN) longitudinal study protocol: Phase 4 (Enrichment) and Phase 5 (Rescan)*)

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2026-05-15

```r
library(ggplot2)
library(ggpubr)
library(gridExtra)
library(grid)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'

## The following object is masked from 'package:ggpubr':
##
##     get_legend
```

```r
library(GGally)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following object is masked from 'package:gridExtra':
##
##     combine

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(patchwork)
```

```
##
## Attaching package: 'patchwork'
```

```
## The following object is masked from 'package:cowplot':
##
##     align_plots
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.1     v stringr   1.6.0
## v lubridate 1.9.5     v tibble    3.3.1
## v purrr     1.2.2     v tidyr     1.3.2
## v readr     2.2.0
```

```
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::combine()   masks gridExtra::combine()
## x dplyr::filter()    masks stats::filter()
## x dplyr::lag()       masks stats::lag()
## x lubridate::stamp() masks cowplot::stamp()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(isotree)
library(lmerTest)
```

```
## Loading required package: lme4
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
##
## Registered S3 method overwritten by 'lme4':
##   method           from
##   na.action.merMod car
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```r
library(irr)
```

```
## Loading required package: lpSolve
```

#Plotting function helper

```r
plot_normality <- function(data, column_name) {
  column_data <- data[[column_name]]
  data_clean <- na.omit(column_data)
    density_estimate <- density(data_clean)
    normal_max <- dnorm(mean(data_clean), mean = mean(data_clean), sd = sd(data_clean))
    ylim_max <- max(density_estimate$y, normal_max)
    hist(data_clean,  main = NA, xlab = column_name, ylab = "Density",
         breaks = 20, col = "skyblue", border = "white", probability = TRUE,
         ylim = c(0, ylim_max * 1.1),
         las = 1)
    lines(density_estimate, col = "blue", lwd = 2, lty = 1)
    curve(dnorm(x, mean = mean(data_clean), sd = sd(data_clean)),
          col = "red", lwd = 2, lty = 2, add = TRUE)
    legend("topright", inset = c(0, -0.2), legend = c("Density", "Normal Distribution"), col = c("blue"
}

plot_violin <- function(data, column_name, group_column) {
  # Remove missing values and prepare the data
  data_clean <- data[!is.na(data[[column_name]]) & !is.na(data[[group_column]]), ]
  data_clean[[column_name]] <- scale(data_clean[[column_name]])
  # Create violin plot with grouping
  ggplot(data_clean, aes_string(x = group_column, y = column_name, fill = group_column)) +
    geom_violin(trim = FALSE, alpha = 0.7) +
    geom_boxplot(width = 0.1, position = position_dodge(0.9), alpha = 0.5 ) +
    scale_fill_manual(
      values = c("0" = "lightblue", "1" = "darkorange")  # Custom colors
    ) +
    ylim(-4, 4) +
    scale_x_discrete(labels = c("0" = "NR", "1" = "R")) +  # Relabel x-axis
    labs(x = NULL, y = column_name) +  # Remove legend title
    theme_minimal() +
    theme(legend.position = "none")  # Remove the legend
}

plot_violin_outlier_suppress <- function(data, column_name, group_column) {
  # Remove missing values and prepare the data
  data_clean <- data[!is.na(data[[column_name]]) & !is.na(data[[group_column]]), ]
  data_clean[[column_name]] <- scale(data_clean[[column_name]])
  # Create violin plot with grouping
  ggplot(data_clean, aes_string(x = group_column, y = column_name, fill = group_column)) +
    geom_violin(trim = FALSE, alpha = 0.7) +
    geom_boxplot(width = 0.1, position = position_dodge(0.9), alpha = 0.5, outlier.shape = NA ) +
    scale_fill_manual(
      values = c("0" = "lightblue", "1" = "darkorange")  # Custom colors
    ) +
    ylim(-3.5, 3.5) +
    scale_x_discrete(labels = c("0" = "NR", "1" = "R")) +  # Relabel x-axis
    labs(x = NULL, y = column_name) +  # Remove legend title
    theme_minimal() +
    theme(legend.position = "none")  # Remove the legend
}
plot_outliers <- function(data, x_var, y_var, x_label, y_label, ymin, ymax, outliers) {
  outlier_CCIDs <- unique(outliers$CC.ID)
```

```r
  data <- data %>%
    mutate(is_outlier = ifelse(CC.ID %in% outlier_CCIDs, TRUE, FALSE))

  ggplot(data, aes(x = .data[[x_var]], y = .data[[y_var]])) +
    geom_point(aes(color = is_outlier, alpha = is_outlier), size = 2, stroke = 1) +
    geom_line(aes(group = CC.ID), lty = 2, alpha = 0.2, color = "black") +
    geom_smooth(
      aes(group = CC.ID, color = is_outlier),
      method = "lm", se = FALSE, size = 0.5, lty = 1
    ) +
    labs(x = x_label, y = y_label) +
    ylim(ymin, ymax) +
    scale_color_manual(
      values = c("TRUE" = "green", "FALSE" = "darkorange"),
      labels = c("TRUE" = "Outlier", "FALSE" = "Non-Outlier")
    ) +
    scale_alpha_manual(values = c("TRUE" = 0.3, "FALSE" = 0.1), guide = "none") +
    theme_minimal() +
    theme(
      plot.title = element_text(hjust = 0.5),
      legend.position = "bottom",
      legend.box = "horizontal"
    ) +
    guides(color = guide_legend(title = NULL))
}
```

Mydata contains ALL cross-sectional and longitudinal data across all phases for 1) Cattell, 2) STW, 3) Memory Imm, 4) Memory Del, 5) Simple Spped (inv RT Simple), 6) Choice Speed (inv RT choice).

```r
mydata <- read.csv("CambridgeCentreForAg-AllCognitiveData_DATA_2026-05-17_1254.csv")
```

#Data wrangling

```r
mydata$phase <- ifelse(grepl("phase_1", mydata$redcap_event_name), "1",
                ifelse(grepl("phase_2", mydata$redcap_event_name), "2",
                ifelse(grepl("phase_3", mydata$redcap_event_name), "3",
                ifelse(grepl("phase_4", mydata$redcap_event_name), "4",
                ifelse(grepl("phase_5", mydata$redcap_event_name), "5",
                ifelse(grepl("participant_info", mydata$redcap_event_name), "999", NA))))))

names(mydata)[names(mydata) == "record_id"] <- "CC.ID"
names(mydata)[names(mydata) == "question_num_years_edu"] <- "education_years"
names(mydata)[names(mydata) == "phase_cog_age"] <- "Age"
names(mydata)[names(mydata) == "cattell_total_score"] <- "Cattell"
names(mydata)[names(mydata) == "stw_n_correct_trials"] <- "STW"
names(mydata)[names(mydata) == "story_memory_immediate_score"] <- "mem_imm"
names(mydata)[names(mydata) == "story_memory_delayed_score"] <- "mem_del"
names(mydata)[names(mydata) == "story_memory_recog_score"] <- "mem_recog"
names(mydata)[names(mydata) == "rt_simple_inv_rt_mean"] <- "simple_speed"
names(mydata)[names(mydata) == "rt_choice_inv_rt_mean"] <- "choice_speed"
names(mydata)[names(mydata) == "phase_cog_online_paper"] <- "format"

mydata$education_years <- as.numeric(mydata$education_years)
```

```r
mydata <- mydata %>%
  group_by(CC.ID) %>%
  mutate(education_years = {
    val <- education_years[phase == 1 & !is.na(education_years)]
    if (length(val) > 0) val[1] else education_years
  }) %>%
  ungroup()

mydata <- mydata %>%
  group_by(CC.ID) %>%
  mutate(across(c(sex, dob, phase_code), ~ {
    val <- .x[phase == 999 & !is.na(.x)]
    if (length(val) > 0) val[1] else .x
  })) %>%
  ungroup()

mydata <- mydata %>% filter(phase != 999 | is.na(phase))

cog_measures <- c("Cattell", "STW", "mem_imm", "mem_del",
                  "simple_speed", "choice_speed")
mydata <- mydata %>% mutate(across(all_of(cog_measures), as.numeric))

mydata$wave <- NA
mydata$wave[mydata$phase == 1] <- 0
mydata$wave[mydata$phase == 2] <- 0
mydata$wave[mydata$phase == 4] <- 1
mydata$wave[mydata$phase == 5] <- 2

mydata$mem_imm[mydata$CC.ID == 620697 & mydata$wave == 2] <- NA #REDCAP ENTRY MISTAKE
mydata$mem_del[mydata$CC.ID == 620697 & mydata$wave == 2] <- NA #REDCAP ENTRY MISTAKE

subject_patterns <- mydata %>%
  mutate(
    participated = as.integer(if_any(all_of(cog_measures), ~ !is.na(.x)))
  ) %>%
  group_by(CC.ID) %>%
  summarize(
    wave0 = as.integer(any(participated == 1 & wave == 0)),
    wave1 = as.integer(any(participated == 1 & wave == 1)),
    wave2 = as.integer(any(participated == 1 & wave == 2)),
    .groups = "drop"
  ) %>%
  mutate(
    pattern = paste0(wave0, wave1, wave2),
    group = case_when(
      pattern == "100"              ~ "G1",
      pattern == "110"              ~ "G2",
      pattern %in% c("111", "101") ~ "G3",
      TRUE                          ~ "Other"
    )
  )

table(subject_patterns$pattern, useNA = "ifany")
```

```
##
##   000   010   100   101   110   111
##    35     3 2016    42   496   110
```

```r
table(subject_patterns$group,   useNA = "ifany")
```

```
##
##     G1     G2     G3 Other
##   2016    496    152    38
```

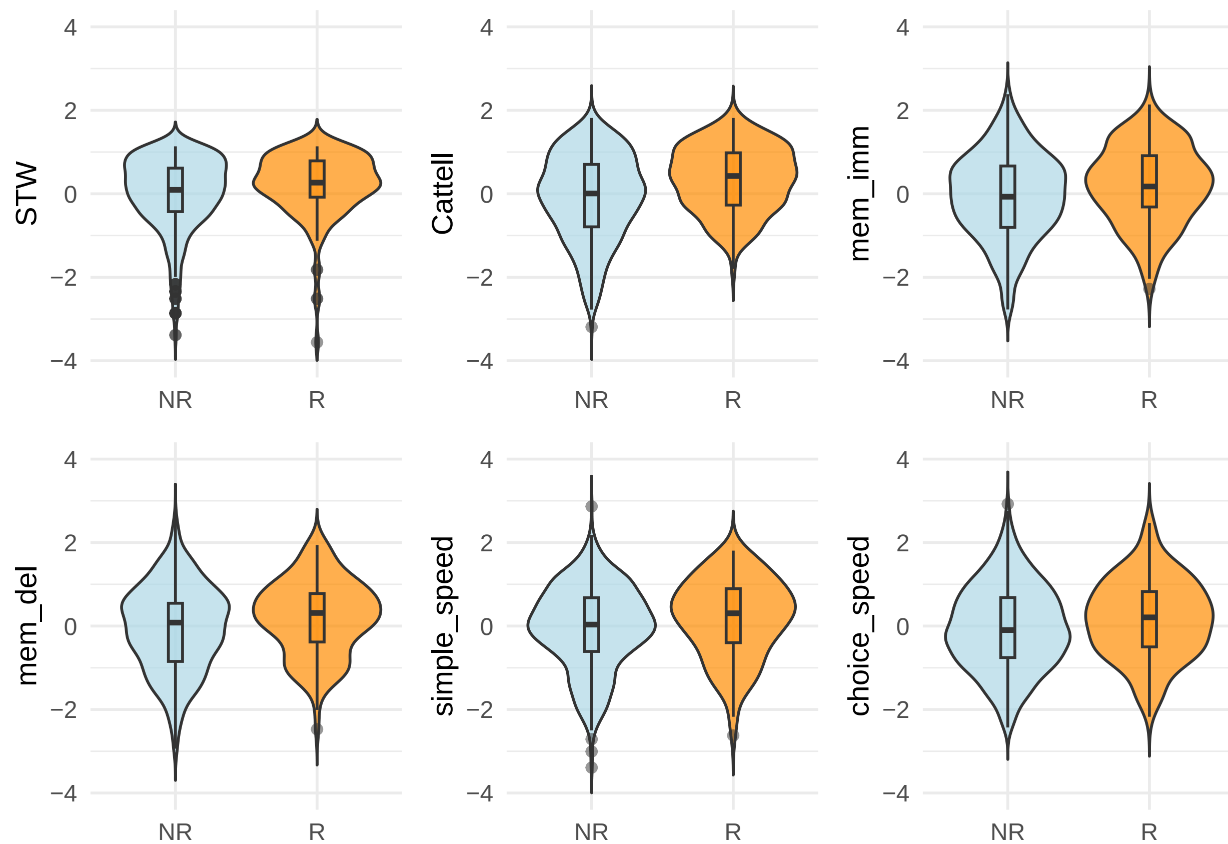
```r
mydata <- merge(mydata, subject_patterns, by = "CC.ID")

missing_ids <- read.csv("missing_record_ids.csv")
names(missing_ids)[names(missing_ids) == "record_id"] <- "CC.ID"
mydata <- mydata[!mydata$CC.ID %in% missing_ids$CC.ID, ]
mydata %>%  group_by(group) %>% summarise(n_unique = n_distinct(CC.ID, na.rm = FALSE))
```

```
## # A tibble: 4 x 2
##   group n_unique
##   <chr>    <int>
## 1 G1        2003
## 2 G2         495
## 3 G3         152
## 4 Other       31
```

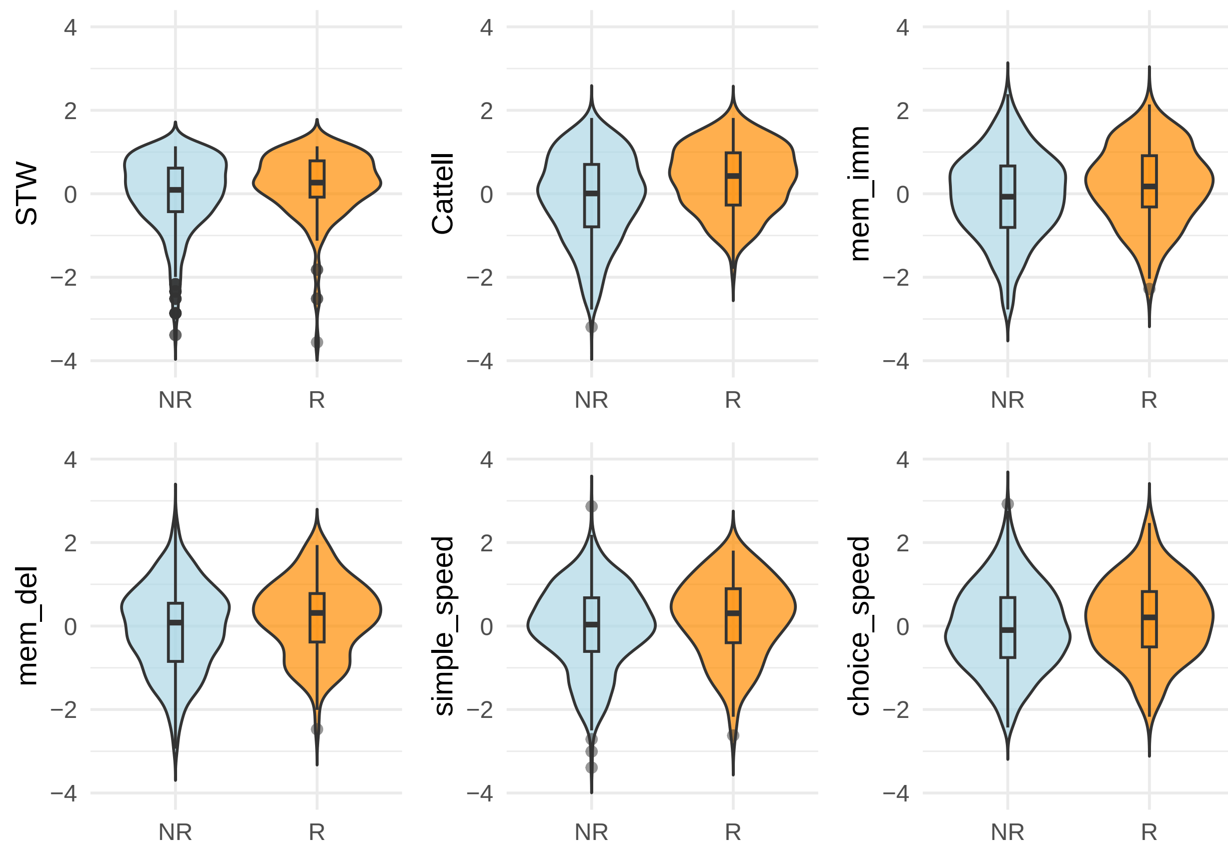
```r
mydata$return_code <- ifelse(mydata$group == "G3", 1, 0)
mydata$return_code <- as.factor(mydata$return_code)

mydata$dob             <- as.Date(mydata$dob)
mydata$phase_cog_date <- as.Date(mydata$phase_cog_date)
mydata$Age             <- round(as.numeric((mydata$phase_cog_date - mydata$dob) / 365.25), 2)
mydata$sex             <- as.factor(mydata$sex)
```

```r
CC700 <- read.csv("CC700_CC.ID.csv")
names(CC700)[names(CC700) == "record_id"] <- "CC.ID"

mydata700 <- mydata[mydata$CC.ID %in% CC700$CC.ID, ]

mydata1_R_NR <- mydata700[mydata700$phase == 1, ]
mydata2_R_NR <- mydata700[mydata700$phase == 2, ]
```

```r
p1 <- plot_violin(mydata1_R_NR, "STW", "return_code")
p2 <- plot_violin(mydata2_R_NR, "Cattell", "return_code")
p3 <- plot_violin(mydata1_R_NR, "mem_imm", "return_code")
p4 <- plot_violin(mydata1_R_NR, "mem_del", "return_code")
p5 <- plot_violin(mydata1_R_NR, "simple_speed", "return_code")
p6 <- plot_violin(mydata1_R_NR, "choice_speed", "return_code")
grid.arrange(p1, p2, p3, p4, p5, p6, ncol = 3)
```




```
#Keep raw values for variables that will be transformed
mydata$STW_raw <- mydata$STW
mydata$STW     <- (mydata$STW_raw)^3

#check normality after transformation
p1 <- plot_violin(mydata1_R_NR, "STW", "return_code")
p2 <- plot_violin(mydata2_R_NR, "Cattell", "return_code")
p3 <- plot_violin(mydata1_R_NR, "mem_imm", "return_code")
p4 <- plot_violin(mydata1_R_NR, "mem_del", "return_code")
p5 <- plot_violin(mydata1_R_NR, "simple_speed", "return_code")
p6 <- plot_violin(mydata1_R_NR, "choice_speed", "return_code")

grid.arrange(p1, p2, p3, p4, p5, p6, ncol = 3)
```



Plot violins without outliers

```
# p1 <- plot_violin_outlier_suppress(mydata1_R_NR, "STW", "return_code")
# p2 <- plot_violin_outlier_suppress(mydata2_R_NR, "Cattell", "return_code")
# p3 <- plot_violin_outlier_suppress(mydata1_R_NR, "mem_imm", "return_code")
# p4 <- plot_violin_outlier_suppress(mydata1_R_NR, "mem_del", "return_code")
# p5 <- plot_violin_outlier_suppress(mydata1_R_NR, "simple_speed", "return_code")
# p6 <- plot_violin_outlier_suppress(mydata1_R_NR, "choice_speed", "return_code")

# grid.arrange(p1, p2, p3, p4, p5, p6, ncol = 3)
```

Phase separation between Phase 1 and Phase 2 is a pratical helper due to Redcap construction. We treat cognitive variables collected in Phase 1 as Phase 2 interchangeably (due to a very short time lag).

```
mydata1_R <- mydata1_R_NR[mydata1_R_NR$group == "G3", ]
mydata2_R <- mydata2_R_NR[mydata2_R_NR$group == "G3", ]

mydata1_NR <- mydata1_R_NR[mydata1_R_NR$group != "G3" & !is.na(mydata1_R_NR$CC.ID), ]
mydata2_NR <- mydata2_R_NR[mydata2_R_NR$group != "G3" & !is.na(mydata2_R_NR$CC.ID), ]
```

```
g3_in_phase1 <- data.frame(CC.ID = unique(mydata$CC.ID[mydata$group == "G3"]))

CCTina <- read.csv("CC.ID_Tina.csv")
setdiff(CCTina$CC.ID, g3_in_phase1$CC.ID)
```

```
## [1] 420435
```

```r
setdiff(g3_in_phase1$CC.ID, mydata1_R$CC.ID)
```

```
## integer(0)
```

AGE AND SEX DISTRIBUTION

```r
summary(mydata$Age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
##   18.49   43.92   64.17   61.11   79.17  102.17    6839
```

```r
table(mydata$sex)         # 2 - female #1 - male
```

```
##
##    1    2
## 4871 6200
```

```r
summary(mydata1_R$Age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   24.00   42.30   56.96   55.01   66.40   83.41
```

```r
summary(mydata1_NR$Age)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NAs
##   18.50   41.46   59.17   57.50   75.50   94.00       1
```

```r
table(mydata1_R$sex)
```

```
##
##  1  2
## 89 63
```

```r
table(mydata1_NR$sex)
```

```
##
##   1   2
## 299 325
```

```r
sex_ratio <- table(mydata1_R_NR$return_code, mydata1_R_NR$sex)
print(sex_ratio)
```

```
##
##       1   2
##   0 299 325
##   1  89  63
```

```r
chisq.test(sex_ratio)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  sex_ratio
## X-squared = 5.1134, df = 1, p-value = 0.02374
```

```r
m_age <- glm(return_code ~ Age, data = mydata1_R_NR, na.action = na.omit, family = binomial)
anova(m_age, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: return_code
##
## Terms added sequentially (first to last)
##
##
##      Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                   774     767.23
## Age   1     2.16       773     765.07   0.1416
```

```r
m_sex <- glm(return_code ~ sex, data = mydata1_R_NR, na.action = na.omit, family = binomial)
anova(m_sex, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: return_code
##
## Terms added sequentially (first to last)
##
##
##      Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                   775     767.67
## sex   1    5.553       774     762.12  0.01845 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
m_edu <- glm(return_code ~ education_years, data = mydata1_R_NR, na.action = na.omit, family = binomial
anova(m_edu, test = "Chisq")
```

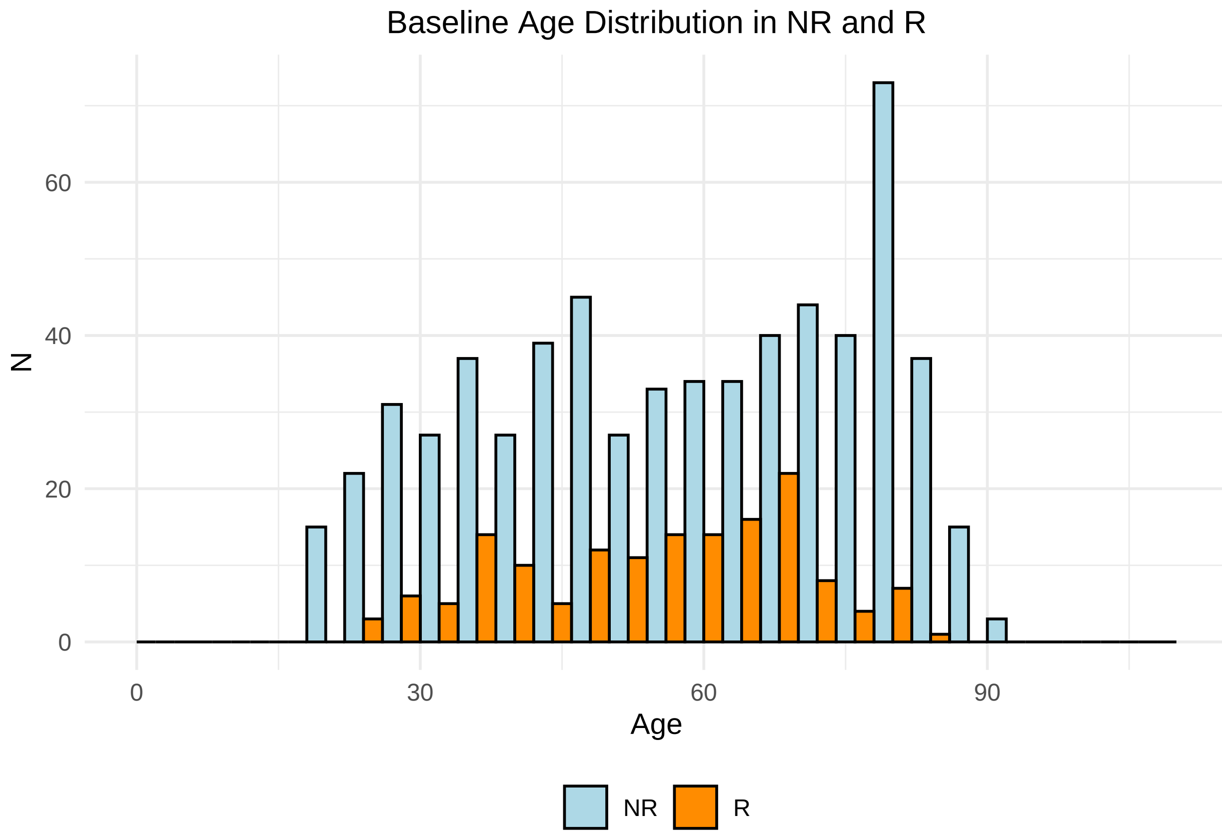
```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: return_code
##
```

```
## Terms added sequentially (first to last)
##
##
##                 Df Deviance Resid. Df Resid. Dev  Pr(>Chi)
## NULL                              773     766.80
## education_years  1   21.511       772     745.28 3.518e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

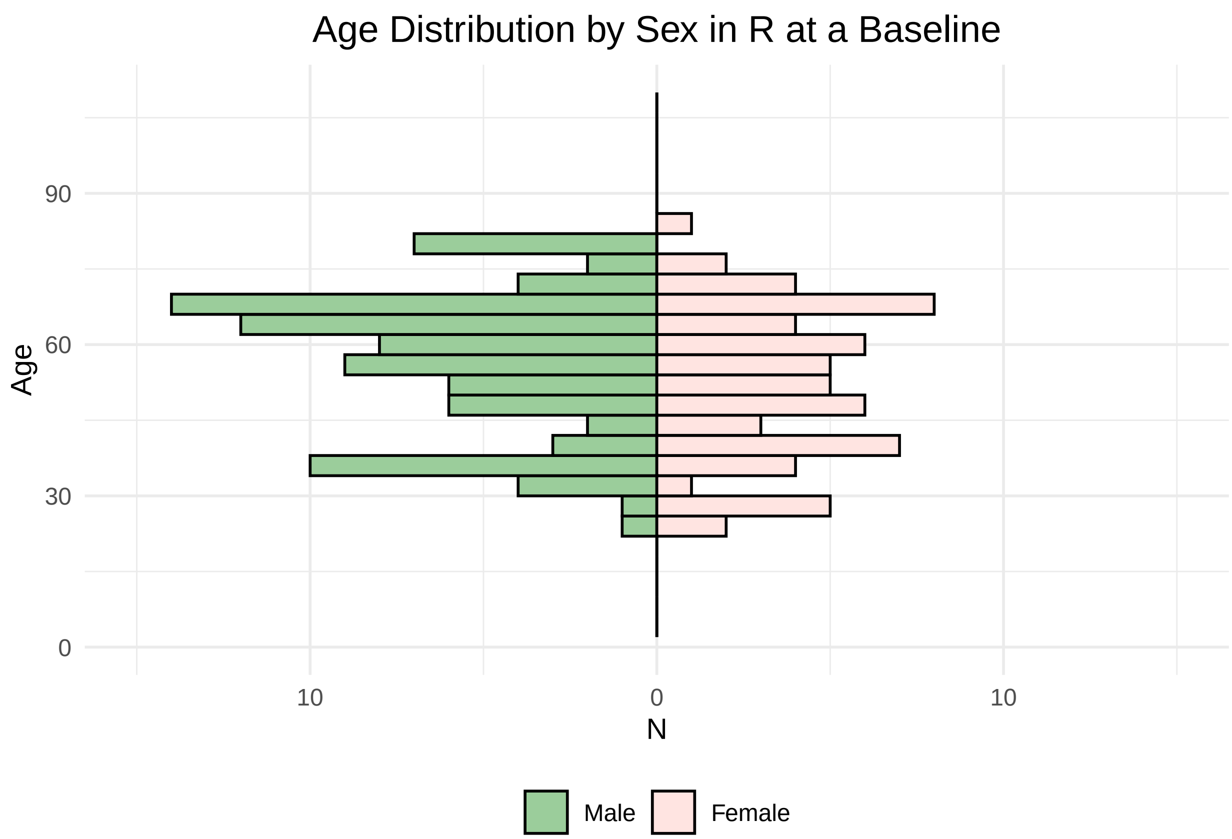
```r
m_full <- glm(return_code ~ sex*Age, data = mydata1_R_NR, na.action = na.omit, family = binomial)
anova(m_full, test = "Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: return_code
##
## Terms added sequentially (first to last)
##
##
##         Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                      774     767.23
## sex      1   5.6385       773     761.59  0.01757 *
## Age      1   2.3894       772     759.21  0.12216
## sex:Age  1   1.1725       771     758.03  0.27890
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
ggplot(mydata1_R_NR, aes(x = Age, fill = return_code)) +
  geom_histogram(binwidth = 4, position = "dodge", color = "black", size = 0.2) +
  labs(x = "Age", y = "N", fill = "Group") +
  ggtitle("Baseline Age Distribution in NR and R") +
  scale_fill_manual(values = c("lightblue", "darkorange"),
                    labels = c("0" = "NR", "1" = "R"), name = NULL) +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, size = 12),
        legend.position = "bottom") +
  xlim(0, 110)
```




```r
mydata1_R <- mydata1_R %>%
  mutate(Age = as.numeric(Age),
         N = ifelse(sex == 1, -1, 1))

ggplot(mydata1_R, aes(x = Age, fill = factor(sex))) +
  geom_histogram(data = filter(mydata1_R, sex == 1), aes(y = -..count..),
                 binwidth = 4, color = "black", size = 0.2, position = "identity") +
  geom_histogram(data = filter(mydata1_R, sex == 2), aes(y = ..count..),
                 binwidth = 4, color = "black", size = 0.2, position = "identity") +
  scale_fill_manual(values = c("#9BCD9B", "#FFE4E1"),
                    labels = c("1" = "Male", "2" = "Female"), name = NULL) +
  labs(x = "Age", y = "N", fill = "sex") +
  ggtitle("Age Distribution by Sex in R at a Baseline") +
  theme_minimal() +
  theme(plot.title = element_text(hjust = 0.5, size = 14),
        legend.position = "bottom") +
  xlim(0, 110) +
  coord_flip() +  # Flip coordinates for a horizontal layout
  scale_y_continuous(labels = abs, limits = c(-15, 15))  # Combine labels and limits
```



EDUCATION

```
outliers_edu    <- boxplot.stats(mydata1_R_NR$education_years)$out
outliers_edu_R  <- boxplot.stats(mydata1_R$education_years)$out
outliers_edu_NR <- boxplot.stats(mydata1_NR$education_years)$out

mydata1_R_NR$education_years[mydata1_R_NR$return_code == 1 & mydata1_R_NR$education_years %in% outliers_
mydata1_R_NR$education_years[mydata1_R_NR$return_code == 0 & mydata1_R_NR$education_years %in% outliers_

wilcox.test(education_years ~ return_code, data = mydata1_R_NR)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  education_years by return_code
## W = 26917, p-value = 1.233e-10
## alternative hypothesis: true location shift is not equal to 0
```

```
summary(mydata1_R$Education)
```

```
## Length  Class   Mode
##      0   NULL   NULL
```

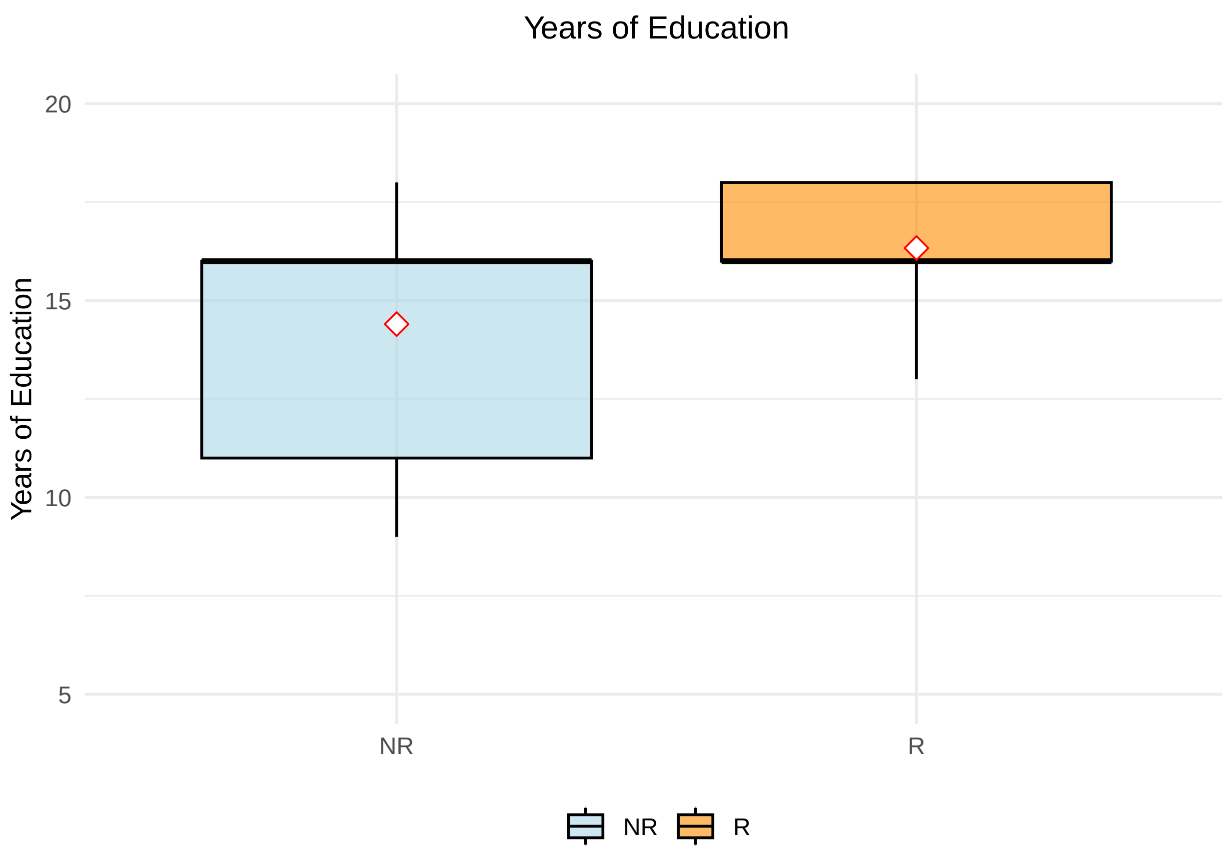
```r
summary(mydata1_NR$Education)
```

```
## Length  Class   Mode
##      0   NULL   NULL
```

```r
ggplot(mydata1_R_NR, aes(x = factor(return_code), y = education_years, fill = factor(return_code))) +
  geom_boxplot(color = "black", alpha = 0.6) +  # Box plot with some transparency
  stat_summary(fun = mean, geom = "point", shape = 23, size = 3, color = "red", fill = "white") +
  labs(title = "Years of Education",
       y = "Years of Education",
       x = NULL) +
  scale_fill_manual(values = c("0" = "lightblue", "1" = "darkorange"),
                    labels = c("0" = "NR", "1" = "R"),
                    name = NULL) +
  scale_x_discrete(labels = c("0" = "NR", "1" = "R")) +
  theme_minimal() +
  theme(
      legend.position = "bottom",
      legend.background = element_rect(fill = "transparent", color = NA),
      plot.title = element_text(hjust = 0.5, vjust = 3, size = 12)
    ) +
  ylim(5, 20)
```

```
## Warning: Removed 22 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 22 rows containing non-finite outside the scale range
## (`stat_summary()`).
```



Education selectivity

```
clean_data <- na.omit(mydata1_R_NR[, c("education_years", "Age", "return_code")])
education_model <- lm(education_years ~ poly(Age,2), data = clean_data)
clean_data$regressed_education <- residuals(education_model, na.rm = TRUE)

r_residuals_ed <- clean_data$regressed_education[clean_data$return_code == "1"]
cc700_residuals_ed <- clean_data$regressed_education

(mean(r_residuals_ed) - (mean(cc700_residuals_ed)))/sd(cc700_residuals_ed)
```

```
## [1] 0.4810646
```

```
(mean(mydata1_R$education_years, na.rm = TRUE) - (mean(mydata1_R_NR$education_years, na.rm = TRUE)))/sd
```

```
## [1] 0.2939932
```

```
clean_data <- na.omit(mydata1_R_NR[, c("education_years", "sex", "return_code", "Age")])
m_education <-lm(education_years ~ return_code * poly(Age, 2), data = clean_data)
summary(m_education)
```

```
##
```

```
## Call:
## lm(formula = education_years ~ return_code * poly(Age, 2), data = clean_data)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -5.8091 -2.1060  0.8379  1.8184  5.6228
##
## Coefficients:
##                             Estimate Std. Error t value Pr(>|t|)
## (Intercept)                  14.4503     0.1079 133.933  < 2e-16 ***
## return_code1                  1.8863     0.3068   6.148 1.27e-09 ***
## poly(Age, 2)1               -17.9054     2.8464  -6.291 5.38e-10 ***
## poly(Age, 2)2                -8.2758     2.8692  -2.884  0.00403 **
## return_code1:poly(Age, 2)1   13.2599     9.1450   1.450  0.14749
## return_code1:poly(Age, 2)2   10.2502     9.9133   1.034  0.30147
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.68 on 748 degrees of freedom
## Multiple R-squared:  0.1239, Adjusted R-squared:  0.118
## F-statistic: 21.15 on 5 and 748 DF,  p-value: < 2.2e-16
```

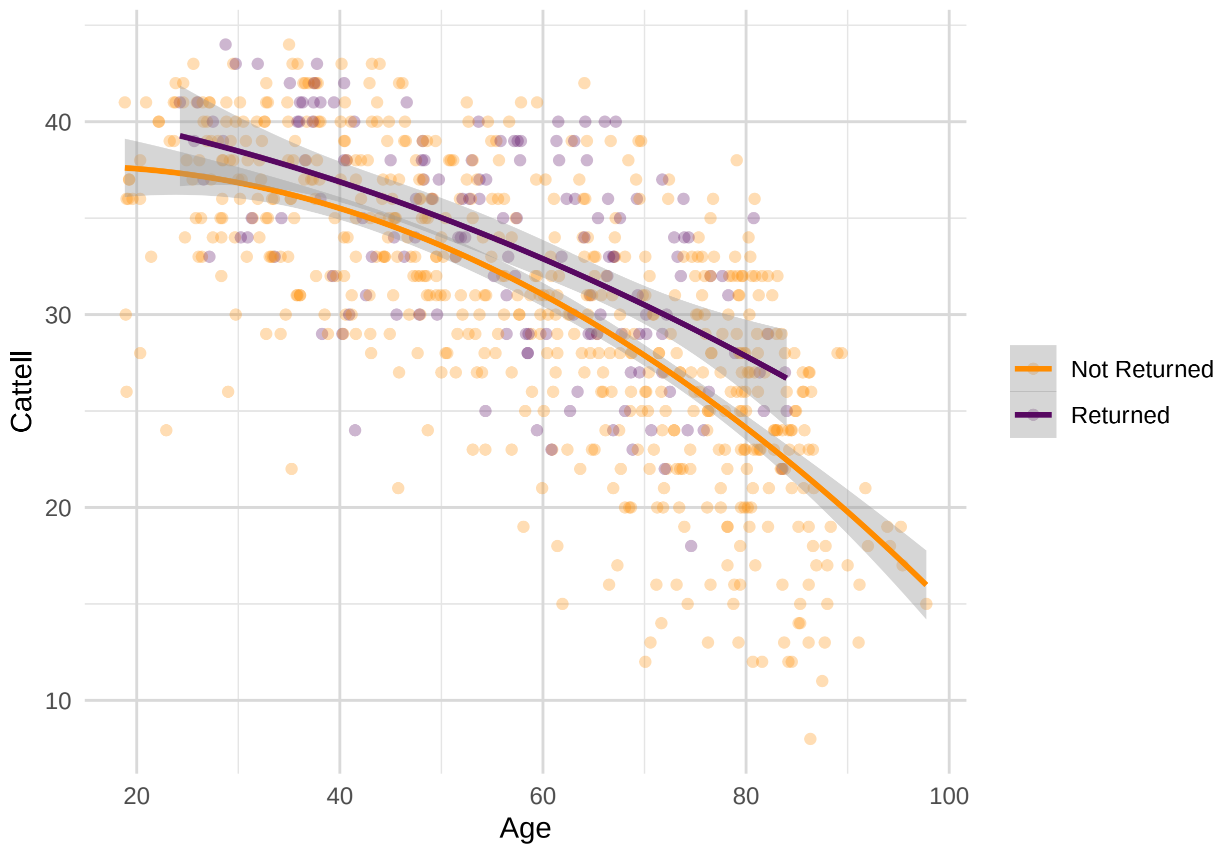
```r
car::Anova(m_education, type=3)
```

```
## Anova Table (Type III tests)
##
## Response: education_years
##                           Sum Sq  Df    F value    Pr(>F)
## (Intercept)               128815   1 17937.9821 < 2.2e-16 ***
## return_code                  271   1    37.8014 1.274e-09 ***
## poly(Age, 2)                 350   2    24.3747 5.557e-11 ***
## return_code:poly(Age, 2)      18   2     1.2389    0.2903
## Residuals                   5371 748
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

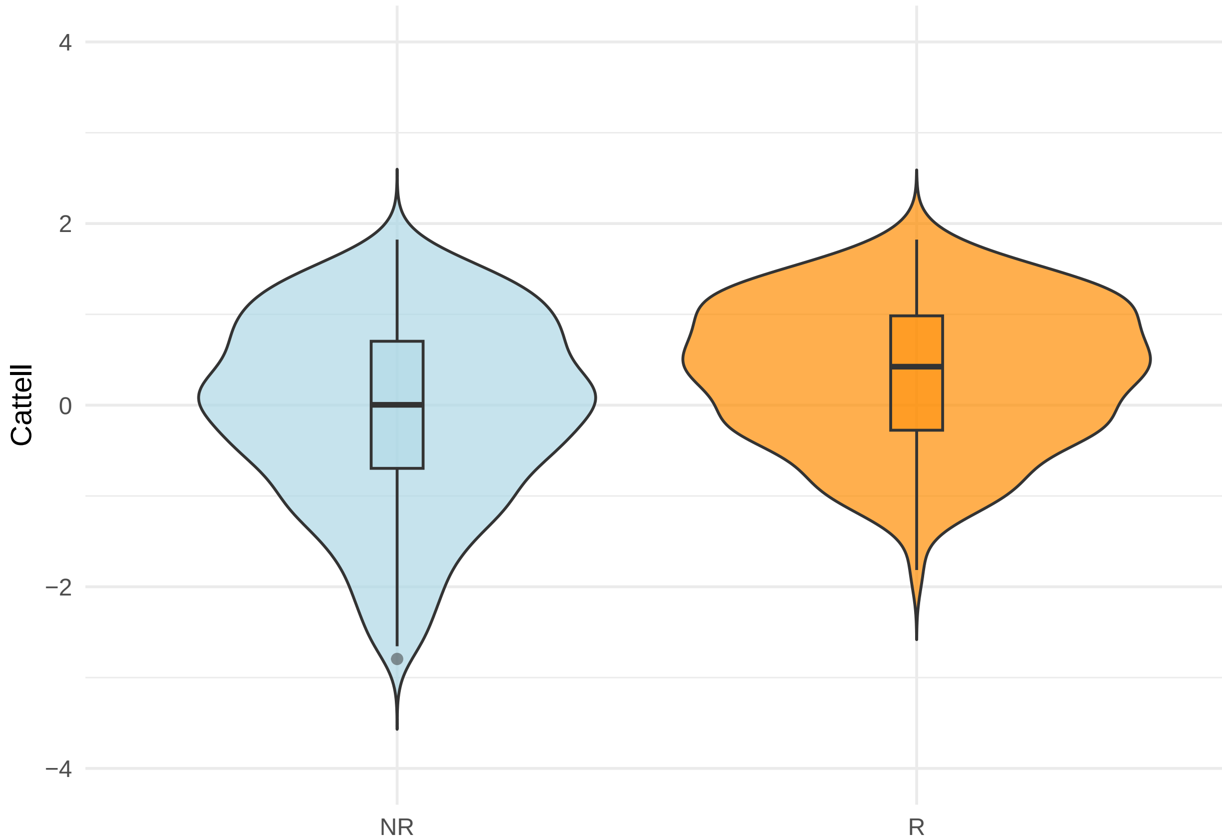
FLUID INTELLIGENCE (CATTELL)

```r
ggplot(mydata2_R_NR, aes(x = Age, y = Cattell, color = return_code)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = , size = 1) +
  labs(x = "Age", y = "Cattell") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    panel.grid.major = element_line(color = "grey85", size = 0.5),
    panel.grid.minor = element_line(color = "grey90", size = 0.25)
  ) +
  scale_color_manual(values = c("darkorange", "#570861"), labels = c("0" = "Not Returned", "1" = "Retur
)
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

## Warning: The 'size' argument of 'element_line()' is deprecated as of ggplot2 3.4.0.
## i Please use the 'linewidth' argument instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

## Warning: Removed 29 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 29 rows containing missing values or values outside the scale range
## ('geom_point()').
```




```
outliers_cattell_R <- boxplot.stats(mydata2_R$Cattell)$out
outliers_cattell_NR <- boxplot.stats(mydata2_NR$Cattell)$out

mydata2_R_NR$Cattell[mydata2_R_NR$return_code == 1 & mydata2_R_NR$Cattell %in% outliers_cattell_R] <- N
mydata2_R_NR$Cattell[mydata2_R_NR$return_code == 0 & mydata2_R_NR$Cattell %in% outliers_cattell_NR] <- N

plot_violin(mydata2_R_NR, "Cattell", "return_code")
```




```r
t.test(mydata2_R$Cattell, mydata2_NR$Cattell, paired = FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  mydata2_R$Cattell and mydata2_NR$Cattell
## t = 5.9178, df = 303.83, p-value = 8.795e-09
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  2.141955 4.276105
## sample estimates:
## mean of x mean of y
##  33.50000  30.29097
```

```r
wilcox.test(Cattell ~ return_code, data = mydata2_R_NR)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  Cattell by return_code
## W = 34450, p-value = 4.427e-06
## alternative hypothesis: true location shift is not equal to 0
```

Cattell Selectivity

```
clean_data <- na.omit(mydata2_R_NR[, c("Cattell", "Age", "education_years", "return_code", "sex")])
clean_data$return_code <- as.factor(clean_data$return_code)
clean_data$sex <- as.factor(clean_data$sex)

cat_model <- lm(Cattell ~ poly(Age,2), data = clean_data)
clean_data$regressed_cattell <- residuals(cat_model, na.rm = TRUE)

returners_residuals <- clean_data$regressed_cattell[clean_data$return_code == "1"]
cc700_residuals <- clean_data$regressed_cattell

#age-orthogonal selectivity statistic
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)
```

```
## [1] 0.3093427
```

```
#age-driven selectivity statistic
(mean(mydata2_R$Cattell, na.rm = TRUE) - (mean(mydata2_R_NR$Cattell, na.rm = TRUE)))/sd(mydata2_R_NR$Ca
```

```
## [1] 0.3537716
```

```
m_cattell <-lm(Cattell ~ return_code * scale(education_years) * poly(Age, 2) * sex, data = clean_data)
summary(m_cattell)
```

```
##
## Call:
## lm(formula = Cattell ~ return_code * scale(education_years) *
##     poly(Age, 2) * sex, data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -15.3818  -3.2177   0.3713   3.4910  11.1433
##
## Coefficients:
##                                                           Estimate Std. Error
## (Intercept)                                                31.1590     0.2859
## return_code1                                                2.1418     0.7323
## scale(education_years)                                      2.3114     0.2930
## poly(Age, 2)1                                            -122.6900     7.3811
## poly(Age, 2)2                                             -25.7948     7.5740
## sex2                                                       -0.8162     0.4024
## return_code1:scale(education_years)                        -1.7056     0.8913
## return_code1:poly(Age, 2)1                                  4.6442    22.0633
## return_code1:poly(Age, 2)2                                 34.5040    22.8786
## scale(education_years):poly(Age, 2)1                        5.7956     7.5658
## scale(education_years):poly(Age, 2)2                       -6.7296     6.9926
## return_code1:sex2                                           0.5353     1.2358
## scale(education_years):sex2                                -0.8776     0.4168
## poly(Age, 2)1:sex2                                          0.3259    10.5699
## poly(Age, 2)2:sex2                                          5.5430    10.6322
```

```
## return_code1:scale(education_years):poly(Age, 2)1        15.0903    27.3630
## return_code1:scale(education_years):poly(Age, 2)2       -26.6348    28.1855
## return_code1:scale(education_years):sex2                  1.3795     1.3120
## return_code1:poly(Age, 2)1:sex2                          -1.3990    36.6257
## return_code1:poly(Age, 2)2:sex2                           3.4391    40.3492
## scale(education_years):poly(Age, 2)1:sex2                 0.5158    11.2743
## scale(education_years):poly(Age, 2)2:sex2                 7.7569    10.7903
## return_code1:scale(education_years):poly(Age, 2)1:sex2   38.1164    41.5858
## return_code1:scale(education_years):poly(Age, 2)2:sex2    5.0286    42.3443
##                                                         t value Pr(>|t|)
## (Intercept)                                             108.988  < 2e-16 ***
## return_code1                                              2.925 0.003553 **
## scale(education_years)                                    7.889 1.13e-14 ***
## poly(Age, 2)1                                           -16.622  < 2e-16 ***
## poly(Age, 2)2                                            -3.406 0.000696 ***
## sex2                                                     -2.028 0.042887 *
## return_code1:scale(education_years)                      -1.914 0.056069 .
## return_code1:poly(Age, 2)1                                0.210 0.833340
## return_code1:poly(Age, 2)2                                1.508 0.131958
## scale(education_years):poly(Age, 2)1                      0.766 0.443914
## scale(education_years):poly(Age, 2)2                     -0.962 0.336178
## return_code1:sex2                                         0.433 0.665038
## scale(education_years):sex2                              -2.106 0.035576 *
## poly(Age, 2)1:sex2                                        0.031 0.975409
## poly(Age, 2)2:sex2                                        0.521 0.602292
## return_code1:scale(education_years):poly(Age, 2)1         0.551 0.581471
## return_code1:scale(education_years):poly(Age, 2)2        -0.945 0.344986
## return_code1:scale(education_years):sex2                  1.051 0.293409
## return_code1:poly(Age, 2)1:sex2                          -0.038 0.969540
## return_code1:poly(Age, 2)2:sex2                           0.085 0.932099
## scale(education_years):poly(Age, 2)1:sex2                 0.046 0.963521
## scale(education_years):poly(Age, 2)2:sex2                 0.719 0.472451
## return_code1:scale(education_years):poly(Age, 2)1:sex2    0.917 0.359672
## return_code1:scale(education_years):poly(Age, 2)2:sex2    0.119 0.905503
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.706 on 721 degrees of freedom
## Multiple R-squared:  0.5815, Adjusted R-squared:  0.5681
## F-statistic: 43.55 on 23 and 721 DF,  p-value: < 2.2e-16
```

```
car::Anova(m_cattell, type=3)
```

```
## Anova Table (Type III tests)
##
## Response: Cattell
##                                                       Sum Sq  Df    F value
## (Intercept)                                           263063   1 11878.4722
## return_code                                              189   1     8.5553
## scale(education_years)                                  1378   1    62.2417
## poly(Age, 2)                                            6451   2   145.6478
## sex                                                       91   1     4.1144
## return_code:scale(education_years)                        81   1     3.6618
## return_code:poly(Age, 2)                                  53   2     1.1910
```

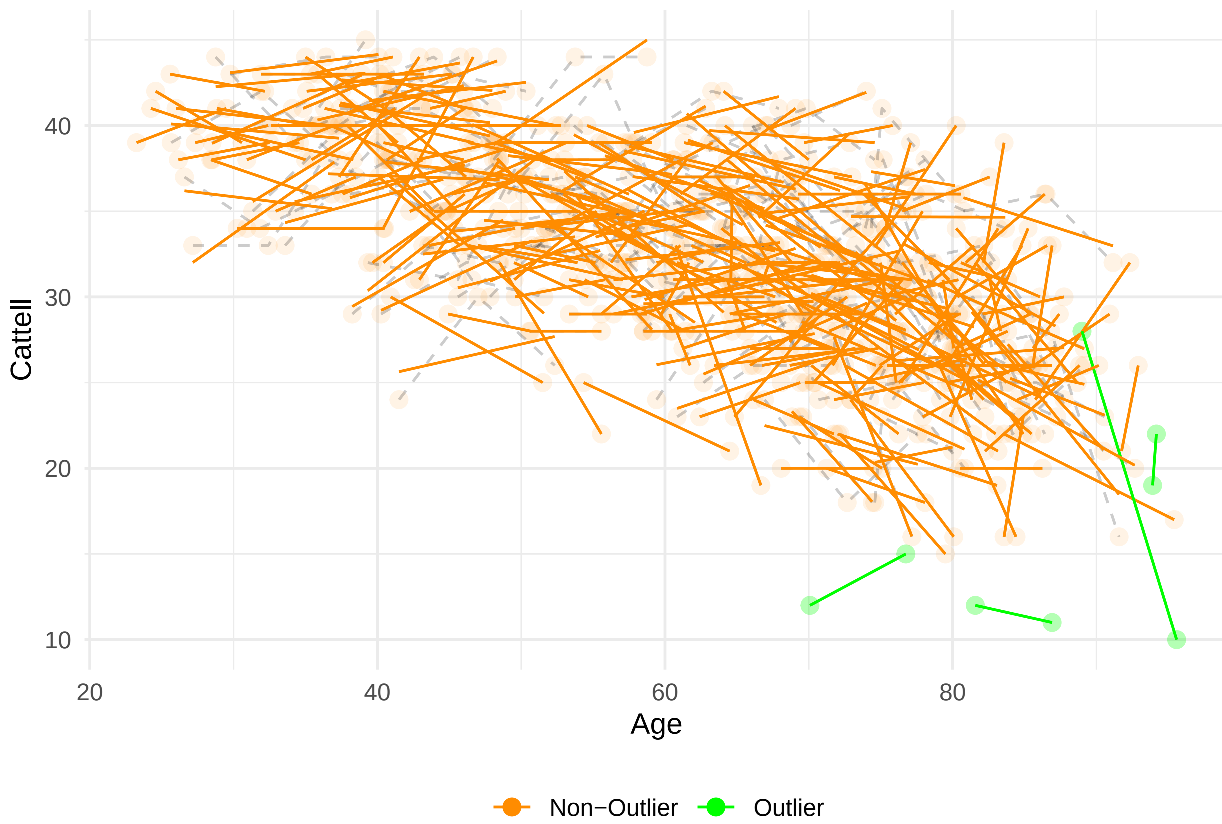
```
## scale(education_years):poly(Age, 2)                      29   2     0.6454
## return_code:sex                                            4   1     0.1876
## scale(education_years):sex                                98   1     4.4340
## poly(Age, 2):sex                                           6   2     0.1362
## return_code:scale(education_years):poly(Age, 2)          25   2     0.5716
## return_code:scale(education_years):sex                   24   1     1.1055
## return_code:poly(Age, 2):sex                               0   2     0.0045
## scale(education_years):poly(Age, 2):sex                   12   2     0.2765
## return_code:scale(education_years):poly(Age, 2):sex     21   2     0.4639
## Residuals                                              15967 721
##                                                            Pr(>F)
## (Intercept)                                          < 2.2e-16 ***
## return_code                                           0.003553 **
## scale(education_years)                               1.128e-14 ***
## poly(Age, 2)                                         < 2.2e-16 ***
## sex                                                   0.042887 *
## return_code:scale(education_years)                    0.056069 .
## return_code:poly(Age, 2)                              0.304501
## scale(education_years):poly(Age, 2)                   0.524757
## return_code:sex                                       0.665038
## scale(education_years):sex                            0.035576 *
## poly(Age, 2):sex                                      0.872671
## return_code:scale(education_years):poly(Age, 2)      0.564876
## return_code:scale(education_years):sex                0.293409
## return_code:poly(Age, 2):sex                          0.995545
## scale(education_years):poly(Age, 2):sex               0.758497
## return_code:scale(education_years):poly(Age, 2):sex  0.629014
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Cattell Attrition (Pattern-mixture) & ICC

```r
df_cattell <- mydata %>%
  filter(group %in% c("G2", "G3"),
         phase %in% c(2, 4, 5)) %>%
  group_by(CC.ID) %>%
  filter(sum(!is.na(Cattell)) >= 2) %>%
  ungroup()

df_cattell <- df_cattell %>%
  group_by(CC.ID) %>%
  mutate(A0 = as.numeric(Age[wave == '0']),
         dA = as.numeric(Age - A0))

df_cattell$prac <- 1
df_cattell$prac[str_detect(df_cattell$wave, "0")] <- 0
df_cattell$prac   <- factor(df_cattell$prac)
df_cattell$format <- factor(df_cattell$format)
df_cattell$format <- ifelse(df_cattell$wave == 0 & is.na(df_cattell$format), 1, df_cattell$format)

CC.ID <- df_cattell$CC.ID[df_cattell$wave == 0]
```

```r
C0    <- df_cattell$Cattell[df_cattell$wave == 0]
C1    <- df_cattell$Cattell[df_cattell$wave == 1]
C2    <- df_cattell$Cattell[df_cattell$wave == 2]
educ  <- df_cattell$education_years[df_cattell$wave == 0]
sex   <- df_cattell$sex[df_cattell$wave == 0]
A0    <- df_cattell$Age[df_cattell$wave == 0]

feature_matrix <- data.frame(CC.ID, C0, C1, C2, A0, sex, educ)

set.seed(987)
iso <- isolation.forest(feature_matrix[, c("C0", "C1", "C2", "A0")], ntrees = 100, sample_size = 128, n

feature_matrix$pred <- predict(iso, feature_matrix[, c("C0", "C1", "C2", "A0")])
outliers <- feature_matrix[feature_matrix$pred > 0.60, c("CC.ID", "C0", "C1", "C2", "A0", "pred")]
outliers_cattell <- outliers

plot_outliers(df_cattell, "Age", "Cattell", "Age", "Cattell", min(df_cattell$Cattell), max(df_cattell$Ca
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 207 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 207 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 203 rows containing missing values or values outside the scale range
## (`geom_line()`).
```




```r
df_cattell$cattell[df_cattell$CC.ID %in% outliers_cattell$CC.ID] <- NA
```

```
## Warning: Unknown or uninitialised column: ‘cattell‘.
```

```r
df_cattell$cattell_scaled[df_cattell$CC.ID %in% outliers_cattell$CC.ID] <- NA
```

```
## Warning: Unknown or uninitialised column: ‘cattell_scaled‘.
```

```r
m_cattell = lmer('scale(Cattell) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + fo
summary(m_cattell)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(Cattell) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + prac +
##    Data: df_cattell
##
## REML criterion at convergence: 1337.2
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.6146 -0.4903  0.0270  0.5123  2.7748
##
## Random effects:
```

```
##  Groups   Name        Variance Std.Dev.
##  CC.ID    (Intercept) 0.3097   0.5565
##  Residual             0.1792   0.4233
## Number of obs: 711, groups:  CC.ID, 306
##
## Fixed effects:
##                                        Estimate Std. Error        df t value
## (Intercept)                            -0.69016    0.16729 617.79306  -4.126
## scale(A0)                              -0.81453    0.09030 655.16421  -9.020
## scale(dA)                              -0.21625    0.06808 454.37324  -3.176
## groupG3                                 0.13899    0.07800 355.99697   1.782
## scale(education_years)                  0.29087    0.06059 298.55232   4.800
## sex2                                   -0.10114    0.07811 299.75128  -1.295
## format                                  0.25644    0.07060 515.74744   3.632
## prac1                                   0.39417    0.09034 451.75313   4.363
## scale(A0):scale(dA)                    -0.10053    0.06765 470.51275  -1.486
## scale(A0):groupG3                       0.09196    0.07837 347.88164   1.173
## scale(dA):groupG3                       0.05690    0.05030 436.19619   1.131
## scale(A0):scale(education_years)        0.08523    0.06474 304.14026   1.316
## scale(A0):sex2                          0.17346    0.07901 302.22272   2.196
## scale(education_years):sex2            -0.10985    0.08774 300.87113  -1.252
## scale(A0):prac1                         0.05668    0.08387 461.69207   0.676
## scale(A0):scale(dA):groupG3            -0.02693    0.04875 446.06757  -0.552
## scale(A0):scale(education_years):sex2  -0.06036    0.08798 304.31224  -0.686
##                                       Pr(>|t|)
## (Intercept)                           4.20e-05 ***
## scale(A0)                              < 2e-16 ***
## scale(dA)                             0.001592 **
## groupG3                               0.075608 .
## scale(education_years)                2.51e-06 ***
## sex2                                  0.196352
## format                                0.000309 ***
## prac1                                 1.59e-05 ***
## scale(A0):scale(dA)                   0.137972
## scale(A0):groupG3                     0.241461
## scale(dA):groupG3                     0.258584
## scale(A0):scale(education_years)      0.189055
## scale(A0):sex2                        0.028886 *
## scale(education_years):sex2           0.211531
## scale(A0):prac1                       0.499508
## scale(A0):scale(dA):groupG3           0.580936
## scale(A0):scale(education_years):sex2 0.493184
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```r
wide <- pivot_wider(df_cattell[df_cattell$group == "G3" & df_cattell$wave %in% c(0, 2),],
  id_cols = CC.ID,
  names_from = wave,
  values_from = Cattell)
```

```r
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

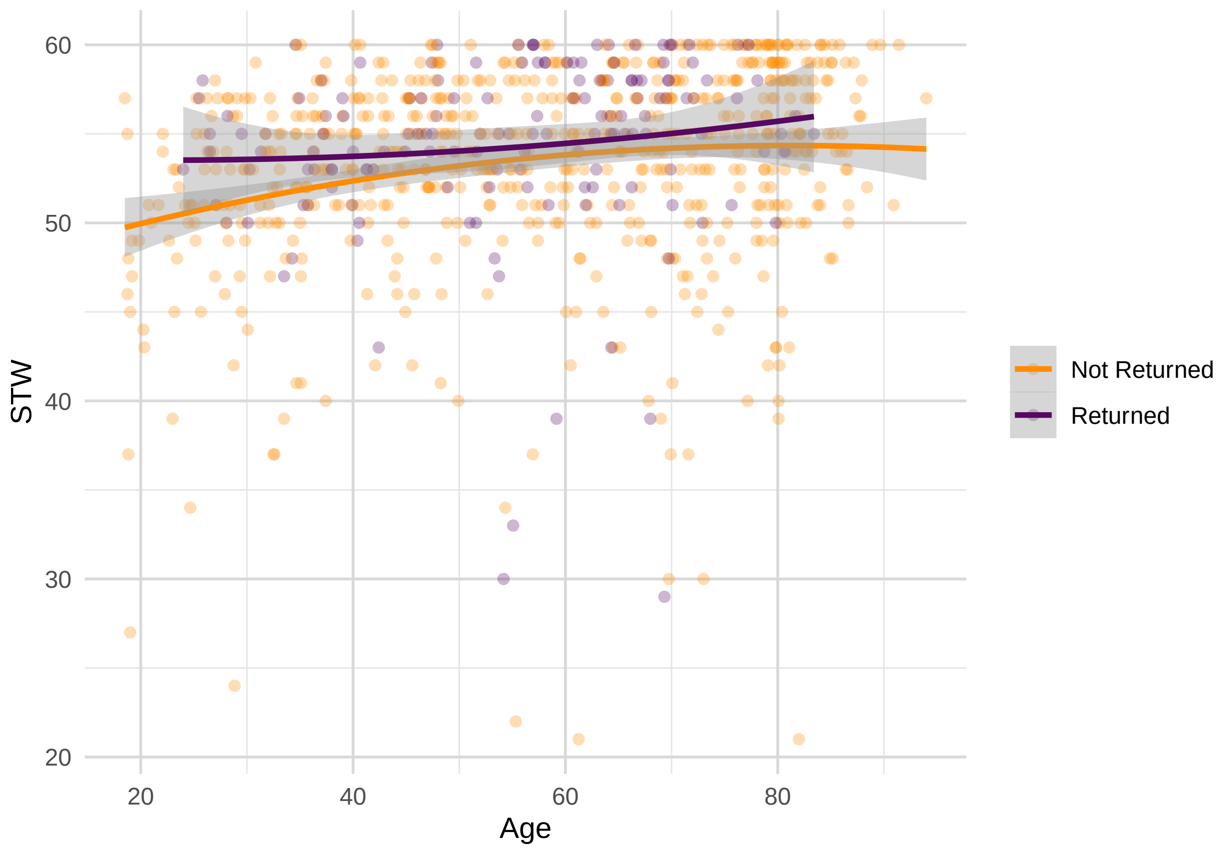
```
##  Single Score Intraclass Correlation
##
##    Model: twoway
##    Type : consistency
##
##    Subjects = 148
##      Raters = 2
##    ICC(C,1) = 0.813
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(147,147) = 9.67 , p = 1.4e-36
##
##  95%-Confidence Interval for ICC Population Values:
##   0.75 < ICC < 0.861
```

STW

```r
ggplot(mydata1_R_NR, aes(x = Age, y = STW, color = return_code)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = , size = 1) +
  labs(x = "Age", y = "STW") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    panel.grid.major = element_line(color = "grey85", size = 0.5),
    panel.grid.minor = element_line(color = "grey90", size = 0.25)
  ) +
  scale_color_manual(values = c("darkorange", "#570861"), labels = c("0" = "Not Returned", "1" = "Retur
                     name = NULL
)
```

```
## Warning: Removed 2 rows containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 2 rows containing missing values or values outside the scale range
## (`geom_point()`).
```




```
sum(is.na(mydata1_R_NR$STW) == TRUE) #N = 2 missing
```

```
## [1] 2
```

```
sum(is.na(mydata1_R$STW) == TRUE)     #N = 0 missing
```

```
## [1] 0
```

```
sum(is.na(mydata1_NR$STW) == TRUE)   #N = 2 missing
```

```
## [1] 2
```

```
outliers_stw_R <- boxplot.stats(mydata1_R$STW)$out
outliers_stw_NR <- boxplot.stats(mydata1_NR$STW)$out

mydata1_R_NR$STW[mydata1_R_NR$return_code == 1 & mydata1_R_NR$STW %in% outliers_stw_R] <- NA
mydata1_R_NR$STW[mydata1_R_NR$return_code == 0 & mydata1_R_NR$STW %in% outliers_stw_NR] <- NA

mydata1_R <- mydata1_R_NR[mydata1_R_NR$phase == 1 & mydata1_R_NR$return_code == 1, ]
mydata1_NR <- mydata1_R_NR[mydata1_R_NR$phase == 1 & mydata1_R_NR$return_code == 0, ]

sum(!is.na(mydata1_R_NR$STW) == TRUE)
```

```
## [1] 739
```

```r
sum(!is.na(mydata1_R$STW) == TRUE)
```

```
## [1] 145
```

```r
sum(!is.na(mydata1_NR$STW) == TRUE)
```

```
## [1] 594
```

```r
t.test(mydata1_R$STW, mydata1_NR$STW, paired = FALSE)
```

```
##
##  Welch Two Sample t-test
##
## data:  mydata1_R$STW and mydata1_NR$STW
## t = 3.555, df = 275.77, p-value = 0.0004445
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  0.5103212 1.7768261
## sample estimates:
## mean of x mean of y
##  55.23448  54.09091
```

```r
wilcox.test(STW ~ return_code, data = mydata1_R_NR)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  STW by return_code
## W = 37227, p-value = 0.01104
## alternative hypothesis: true location shift is not equal to 0
```

STW Selectivity

```r
clean_data <- na.omit(mydata1_R_NR[, c("STW", "Age", "education_years", "return_code", "sex")])
clean_data$return_code <- as.factor(clean_data$return_code)
clean_data$sex <- as.factor(clean_data$sex)

stw_model <- lm(STW ~ poly(Age,2), data = clean_data)
clean_data$regressed_stw <- residuals(stw_model, na.rm = TRUE)

returners_residuals <- clean_data$regressed_stw[clean_data$return_code == "1"]
cc700_residuals     <- clean_data$regressed_stw

(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)
```

```
## [1] 0.311835
```

```r
(mean(mydata1_R$STW, na.rm = TRUE) - (mean(mydata1_R_NR$STW, na.rm = TRUE)))/sd(mydata1_R_NR$STW, na.rm
```

```
## [1] 0.2249765
```

```r
m_stw <-lm(STW ~ return_code*poly(Age,2)*scale(education_years)*sex , data = clean_data)
summary(m_stw)
```

```
##
## Call:
## lm(formula = STW ~ return_code * poly(Age, 2) * scale(education_years) *
##     sex, data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -12.5275  -2.0469   0.3487   2.3522   8.3554
##
## Coefficients:
##                                                       Estimate Std. Error
## (Intercept)                                           54.50134    0.21110
## return_code1                                           0.70381    0.78619
## poly(Age, 2)1                                         27.93620    5.41218
## poly(Age, 2)2                                          3.93726    5.63742
## scale(education_years)                                 2.01658    0.21643
## sex2                                                  -0.35427    0.29440
## return_code1:poly(Age, 2)1                             0.53033   24.18971
## return_code1:poly(Age, 2)2                           -17.12334   23.58383
## return_code1:scale(education_years)                   -1.42445    1.15857
## poly(Age, 2)1:scale(education_years)                  -0.49354    5.60256
## poly(Age, 2)2:scale(education_years)                   2.86639    5.20826
## return_code1:sex2                                      1.44265    1.36156
## poly(Age, 2)1:sex2                                    21.49057    7.63932
## poly(Age, 2)2:sex2                                   -11.11974    7.72740
## scale(education_years):sex2                           -0.09764    0.30257
## return_code1:poly(Age, 2)1:scale(education_years)      4.16587   35.46805
## return_code1:poly(Age, 2)2:scale(education_years)    -30.27757   35.78720
## return_code1:poly(Age, 2)1:sex2                        5.82732   43.36247
## return_code1:poly(Age, 2)2:sex2                       52.55800   45.79383
## return_code1:scale(education_years):sex2              -0.07067    2.04189
## poly(Age, 2)1:scale(education_years):sex2              9.31354    8.15921
## poly(Age, 2)2:scale(education_years):sex2             -7.25977    7.68040
## return_code1:poly(Age, 2)1:scale(education_years):sex2   7.93795   66.57079
## return_code1:poly(Age, 2)2:scale(education_years):sex2  35.39475   65.53125
##                                                      t value Pr(>|t|)
## (Intercept)                                          258.173  < 2e-16 ***
## return_code1                                           0.895  0.37098
## poly(Age, 2)1                                          5.162 3.19e-07 ***
## poly(Age, 2)2                                          0.698  0.48515
## scale(education_years)                                 9.318  < 2e-16 ***
## sex2                                                  -1.203  0.22924
## return_code1:poly(Age, 2)1                             0.022  0.98251
## return_code1:poly(Age, 2)2                            -0.726  0.46804
## return_code1:scale(education_years)                   -1.229  0.21930
```

```
## poly(Age, 2)1:scale(education_years)                       -0.088  0.92983
## poly(Age, 2)2:scale(education_years)                        0.550  0.58225
## return_code1:sex2                                           1.060  0.28971
## poly(Age, 2)1:sex2                                          2.813  0.00504 **
## poly(Age, 2)2:sex2                                         -1.439  0.15060
## scale(education_years):sex2                                -0.323  0.74703
## return_code1:poly(Age, 2)1:scale(education_years)           0.117  0.90653
## return_code1:poly(Age, 2)2:scale(education_years)          -0.846  0.39782
## return_code1:poly(Age, 2)1:sex2                             0.134  0.89314
## return_code1:poly(Age, 2)2:sex2                             1.148  0.25148
## return_code1:scale(education_years):sex2                   -0.035  0.97240
## poly(Age, 2)1:scale(education_years):sex2                   1.141  0.25406
## poly(Age, 2)2:scale(education_years):sex2                  -0.945  0.34487
## return_code1:poly(Age, 2)1:scale(education_years):sex2   0.119  0.90512
## return_code1:poly(Age, 2)2:scale(education_years):sex2   0.540  0.58929
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.456 on 698 degrees of freedom
## Multiple R-squared:  0.3129, Adjusted R-squared:  0.2903
## F-statistic: 13.82 on 23 and 698 DF,  p-value: < 2.2e-16
```

```r
car::Anova(m_stw, type=3)
```

```
## Anova Table (Type III tests)
##
## Response: STW
##                                                 Sum Sq  Df    F value
## (Intercept)                                     796155   1 66653.3928
## return_code                                         10   1     0.8014
## poly(Age, 2)                                       328   2    13.7180
## scale(education_years)                            1037   1    86.8167
## sex                                                 17   1     1.4481
## return_code:poly(Age, 2)                             7   2     0.2789
## return_code:scale(education_years)                  18   1     1.5117
## poly(Age, 2):scale(education_years)                  4   2     0.1519
## return_code:sex                                     13   1     1.1227
## poly(Age, 2):sex                                   120   2     5.0045
## scale(education_years):sex                           1   1     0.1041
## return_code:poly(Age, 2):scale(education_years)      9   2     0.3586
## return_code:poly(Age, 2):sex                        16   2     0.6857
## return_code:scale(education_years):sex               0   1     0.0012
## poly(Age, 2):scale(education_years):sex             22   2     0.9047
## return_code:poly(Age, 2):scale(education_years):sex      4   2     0.1529
## Residuals                                         8337 698
##                                                    Pr(>F)
## (Intercept)                                     < 2.2e-16 ***
## return_code                                       0.37098
## poly(Age, 2)                                    1.434e-06 ***
## scale(education_years)                          < 2.2e-16 ***
## sex                                               0.22924
## return_code:poly(Age, 2)                          0.75673
## return_code:scale(education_years)                0.21930
## poly(Age, 2):scale(education_years)               0.85913
```

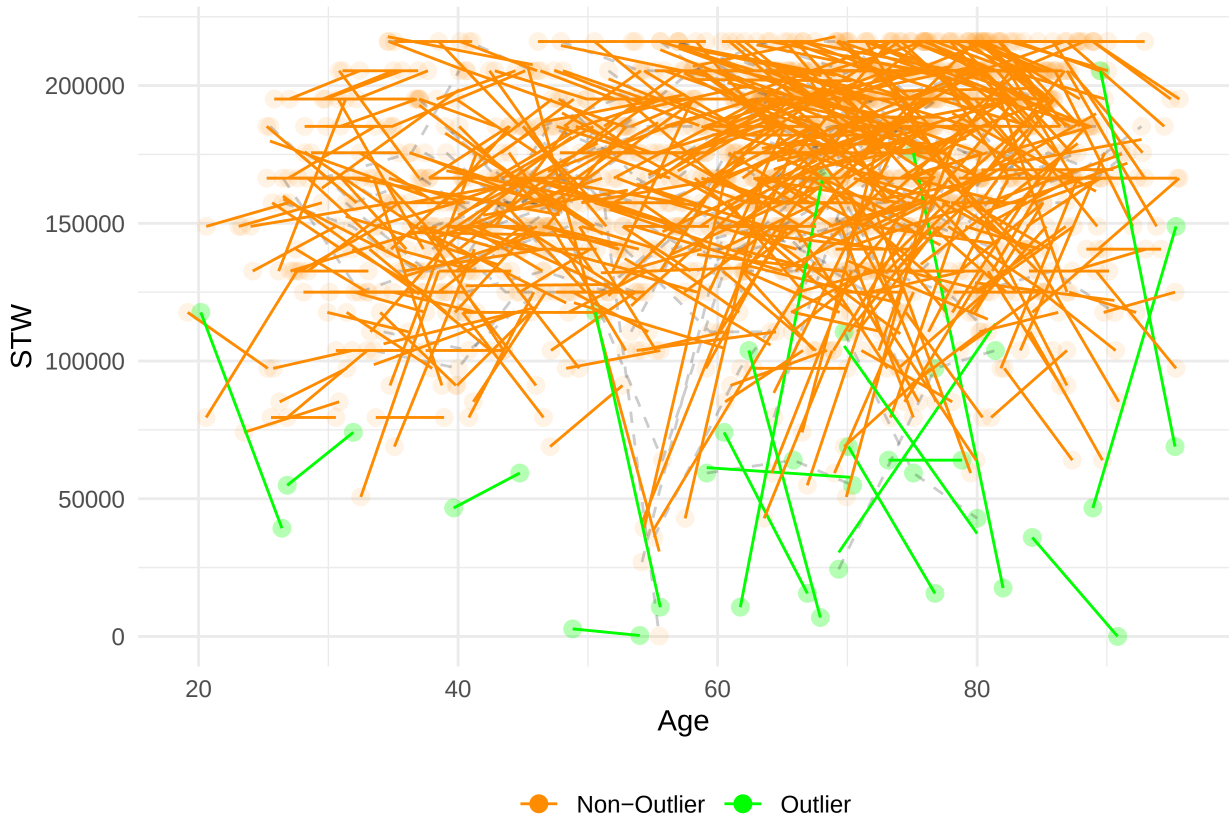
```
## return_code:sex                                          0.28971
## poly(Age, 2):sex                                         0.00695 **
## scale(education_years):sex                               0.74703
## return_code:poly(Age, 2):scale(education_years)          0.69877
## return_code:poly(Age, 2):sex                             0.50408
## return_code:scale(education_years):sex                   0.97240
## poly(Age, 2):scale(education_years):sex                  0.40512
## return_code:poly(Age, 2):scale(education_years):sex      0.85822
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

STW Attrition and ICC

```
df_stw <- mydata %>%
  filter(group %in% c("G2", "G3"),
         phase %in% c(1, 4, 5)) %>%
  group_by(CC.ID) %>%
  filter(sum(!is.na(STW)) >= 2) %>%
  ungroup()

df_stw <- df_stw %>%
  group_by(CC.ID) %>%
  mutate(A0 = as.numeric(Age[wave == '0']),
         dA = as.numeric(Age - A0))

df_stw$prac <- 1
df_stw$prac[str_detect(df_stw$wave, "0")] <- 0
df_stw$prac   <- factor(df_stw$prac)
df_stw$format <- factor(df_stw$format)
df_stw$format <- ifelse(df_stw$wave == 0 & is.na(df_stw$format), 1, df_stw$format)

CC.ID <- df_stw$CC.ID[df_stw$wave == 0]
C0    <- df_stw$STW[df_stw$wave == 0]
C1    <- df_stw$STW[df_stw$wave == 1]
C2    <- df_stw$STW[df_stw$wave == 2]
educ  <- df_stw$education_years[df_stw$wave == 0]
sex   <- df_stw$sex[df_stw$wave == 0]
A0    <- df_stw$Age[df_stw$wave == 0]

feature_matrix <- data.frame(CC.ID, C0, C1, C2, A0, sex, educ)

set.seed(987)
iso <- isolation.forest(feature_matrix[, c("C0", "C1", "C2", "A0")], ntrees = 100, sample_size = 128, n

feature_matrix$pred <- predict(iso, feature_matrix[, c("C0", "C1", "C2", "A0")])
outliers <- feature_matrix[feature_matrix$pred > 0.60, c("CC.ID", "C0", "C1", "C2", "A0", "pred")]
outliers_stw <- outliers

plot_outliers(df_stw, "Age", "STW", "Age", "STW", min(df_stw$STW), max(df_stw$STW), outliers = outliers_
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

```
## Warning: Removed 524 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 524 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 520 rows containing missing values or values outside the scale range
## (`geom_line()`).
```




```r
df_stw$STW[df_stw$CC.ID %in% outliers_stw$CC.ID] <- NA

m_STW = lmer('scale(STW) ~ scale(AO)*scale(dA)*group + scale(AO)*scale(education_years)*sex + format + p
summary(m_STW)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(STW) ~ scale(AO)*scale(dA)*group + scale(AO)*scale(education_years)*sex + format + prac + scal
##    Data: df_stw
##
## REML criterion at convergence: 3048
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -6.2100 -0.4163  0.0704  0.5040  2.9892
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  CC.ID    (Intercept) 0.4236   0.6508
##  Residual             0.3052   0.5525
## Number of obs: 1308, groups:  CC.ID, 606
##
## Fixed effects:
##                                         Estimate Std. Error         df t value
## (Intercept)                            1.234e-01  1.156e-01  1.131e+03   1.067
## scale(A0)                             -3.769e-03  1.087e-01  1.170e+03  -0.035
## scale(dA)                             -1.777e-03  8.594e-02  7.869e+02  -0.021
## groupG3                                8.531e-02  7.958e-02  6.905e+02   1.072
## scale(education_years)                 4.191e-01  5.630e-02  5.851e+02   7.444
## sex2                                  -3.826e-02  6.552e-02  5.915e+02  -0.584
## format                                -1.241e-01  6.526e-02  9.746e+02  -1.902
## prac1                                  9.557e-02  1.389e-01  8.081e+02   0.688
## scale(A0):scale(dA)                   -1.413e-01  9.610e-02  7.997e+02  -1.471
## scale(A0):groupG3                      1.014e-01  8.691e-02  6.868e+02   1.166
## scale(dA):groupG3                     -2.751e-02  5.018e-02  7.576e+02  -0.548
## scale(A0):scale(education_years)       1.558e-01  6.210e-02  5.953e+02   2.509
## scale(A0):sex2                         3.281e-01  6.698e-02  5.976e+02   4.899
## scale(education_years):sex2            4.078e-02  7.314e-02  5.881e+02   0.558
## scale(A0):prac1                        2.043e-01  1.492e-01  8.023e+02   1.369
## scale(A0):scale(dA):groupG3            8.916e-02  5.359e-02  7.603e+02   1.664
## scale(A0):scale(education_years):sex2 -1.135e-01  7.746e-02  5.976e+02  -1.466
##                                       Pr(>|t|)
## (Intercept)                             0.2860
## scale(A0)                               0.9724
## scale(dA)                               0.9835
## groupG3                                 0.2841
## scale(education_years)                3.50e-13 ***
## sex2                                    0.5595
## format                                  0.0575 .
## prac1                                   0.4915
## scale(A0):scale(dA)                     0.1418
## scale(A0):groupG3                       0.2439
## scale(dA):groupG3                       0.5836
## scale(A0):scale(education_years)        0.0124 *
## scale(A0):sex2                        1.24e-06 ***
## scale(education_years):sex2             0.5774
## scale(A0):prac1                         0.1713
## scale(A0):scale(dA):groupG3             0.0966 .
## scale(A0):scale(education_years):sex2   0.1432
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```r
wide <- pivot_wider(df_stw[df_stw$group == "G3" & df_stw$wave %in% c(0, 2),],
  id_cols = CC.ID,
  names_from = wave,
  values_from = STW)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
##  Single Score Intraclass Correlation
##
##    Model: twoway
##    Type : consistency
##
##    Subjects = 143
##      Raters = 2
##    ICC(C,1) = 0.551
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(142,142) = 3.45 , p = 4.32e-13
##
##  95%-Confidence Interval for ICC Population Values:
##   0.425 < ICC < 0.655
```

```r
#sensitivity check
df_sub <- df_stw[df_stw$group == "G3", ]
mod <- lmer(STW ~ 1 + (1 | CC.ID), data = df_sub)
vc <- as.data.frame(VarCorr(mod))
ICC_C <- vc$vcov[vc$grp == "CC.ID"] /
         (vc$vcov[vc$grp == "CC.ID"] + vc$vcov[vc$grp == "Residual"])
ICC_C
```

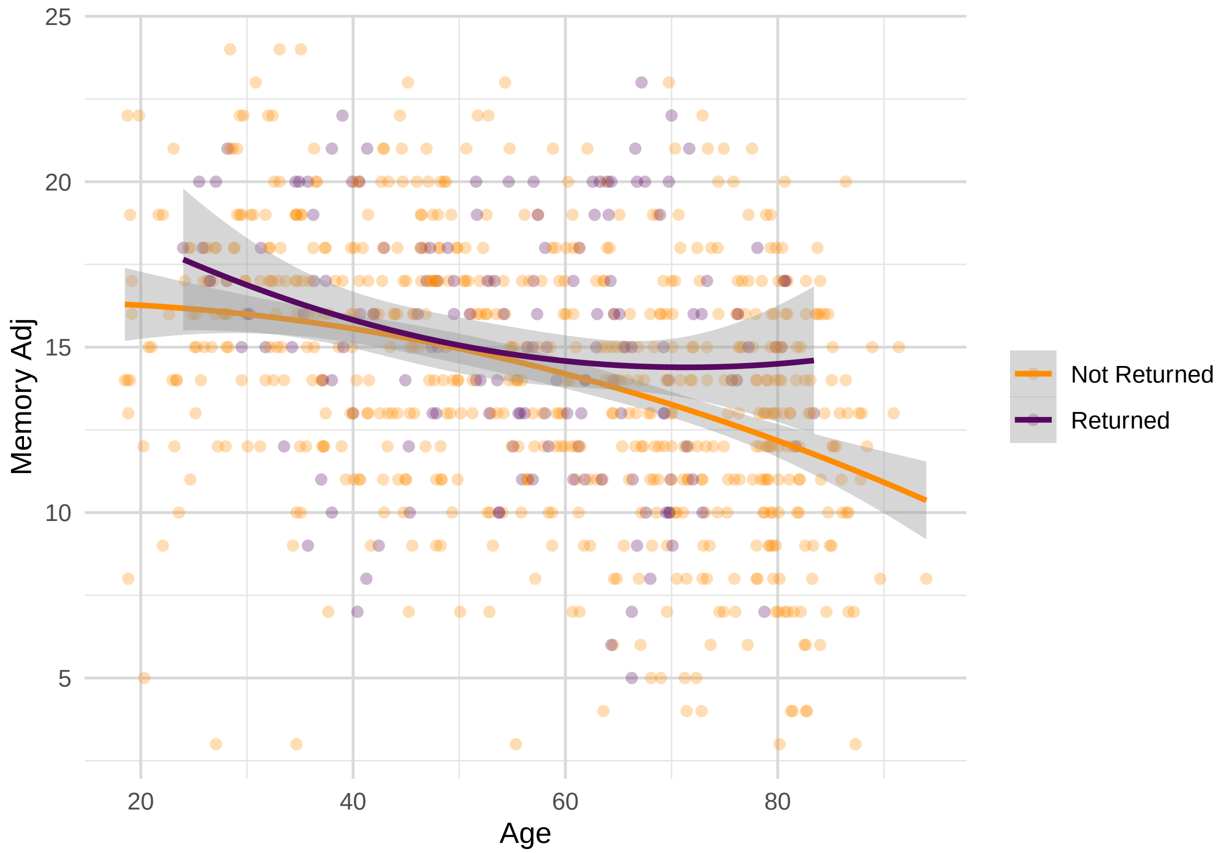
```
## [1] 0.6423334
```

VERBAL EPISODIC MEMORY

```r
ggplot(mydata1_R_NR, aes(x = Age, y = mem_imm, color = return_code)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = TRUE , size = 1) +
  labs(x = "Age", y = "Memory Adj") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    panel.grid.major = element_line(color = "grey85", size = 0.5),
    panel.grid.minor = element_line(color = "grey90", size = 0.25)
  ) +
  scale_color_manual(values = c("darkorange", "#570861"), labels = c("0" = "Not Returned", "1" = "Retur
                     name = NULL
)
```

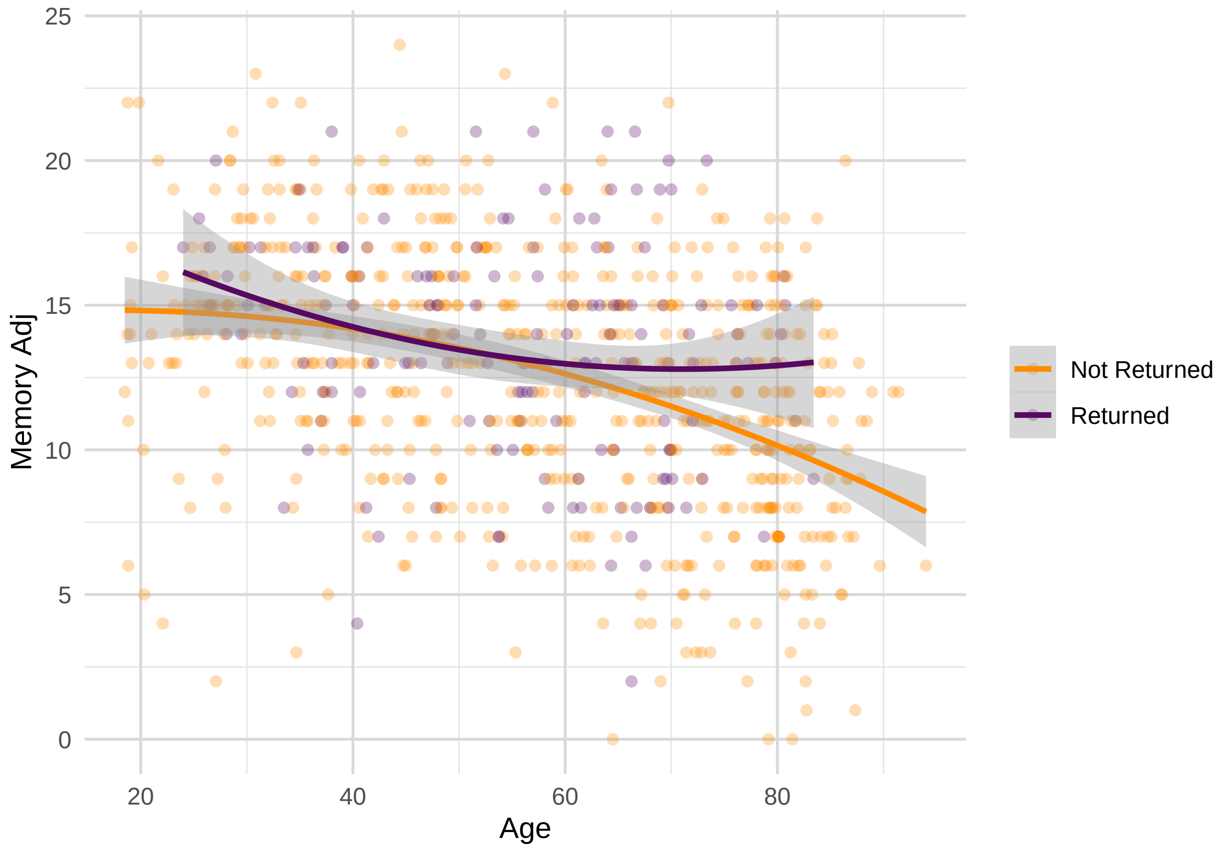
```
## Warning: Removed 1 row containing non-finite outside the scale range
## (`stat_smooth()`).
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
ggplot(mydata1_R_NR, aes(x = Age, y = mem_del, color = return_code)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", formula = y ~ poly(x, 2), se = TRUE , size = 1) +
  labs(x = "Age", y = "Memory Adj") +
  theme_minimal() +
  theme(
    plot.title = element_text(hjust = 0.5, size = 14),
    plot.subtitle = element_text(hjust = 0.5, size = 12),
    panel.grid.major = element_line(color = "grey85", size = 0.5),
    panel.grid.minor = element_line(color = "grey90", size = 0.25)
  ) +
  scale_color_manual(values = c("darkorange", "#570861"), labels = c("0" = "Not Returned", "1" = "Retur
                     name = NULL
)
```

```
## Warning: Removed 1 row containing non-finite outside the scale range (`stat_smooth()`).
## Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```




```
outliers_mem_imm_R  <- boxplot.stats(mydata1_R$mem_imm)$out
outliers_mem_imm_NR <- boxplot.stats(mydata1_NR$mem_imm)$out

mydata1_R_NR$mem_imm[mydata1_R_NR$return_code == 1 & mydata1_R_NR$mem_imm %in% outliers_mem_imm_R] <- NA
mydata1_R_NR$mem_imm[mydata1_R_NR$return_code == 0 & mydata1_R_NR$mem_imm %in% outliers_mem_imm_NR] <- N

outliers_mem_del_R  <- boxplot.stats(mydata1_R$mem_del)$out
outliers_mem_del_NR <- boxplot.stats(mydata1_NR$mem_del)$out

mydata1_R_NR$mem_del[mydata1_R_NR$return_code == 1 & mydata1_R_NR$mem_del %in% outliers_mem_del_R] <- NA
mydata1_R_NR$mem_del[mydata1_R_NR$return_code == 0 & mydata1_R_NR$mem_del %in% outliers_mem_del_NR] <- N

wilcox.test(mem_imm ~ return_code, data = mydata1_R_NR)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  mem_imm by return_code
## W = 40134, p-value = 0.004983
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(mem_del ~ return_code, data = mydata1_R_NR)
```

```
##
```

```
##  Wilcoxon rank sum test with continuity correction
##
## data:  mem_del by return_code
## W = 40080, p-value = 0.004659
## alternative hypothesis: true location shift is not equal to 0
```

Memory Selectivity

```
#selectivity age-driven
(mean(mydata1_R$mem_imm, na.rm = TRUE) - (mean(mydata1_R_NR$mem_imm, na.rm = TRUE)))/sd(mydata1_R_NR$mem
```

```
## [1] 0.1973288
```

```
(mean(mydata1_R$mem_del, na.rm = TRUE) - (mean(mydata1_R_NR$mem_del, na.rm = TRUE)))/sd(mydata1_R_NR$mem
```

```
## [1] 0.1984596
```

```
# selectivity age-orthogonal (and education-adjusted) for Imm
clean_data <- na.omit(mydata1_R_NR[, c("mem_imm", "Age", "education_years", "return_code")])
imm_model <- lm(mem_imm ~ poly(Age,2), data = clean_data)
clean_data$regressed_imm <- residuals(imm_model, na.rm = TRUE)

returners_residuals <- clean_data$regressed_imm[clean_data$return_code == "1"]
cc700_residuals     <- clean_data$regressed_imm
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)
```

```
## [1] 0.1965595
```

```
m_mem <-lm(mem_imm ~ return_code*poly(Age,2)*scale(education_years) , data = clean_data)
summary(m_mem)
```

```
##
## Call:
## lm(formula = mem_imm ~ return_code * poly(Age, 2) * scale(education_years),
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -14.3716  -2.3997   0.2003   2.4571  11.1752
##
## Coefficients:
##                                                     Estimate Std. Error t value
## (Intercept)                                          14.1976     0.1540  92.184
## return_code1                                          1.0513     0.6698   1.570
## poly(Age, 2)1                                       -34.8789     4.0745  -8.560
## poly(Age, 2)2                                        -5.2299     4.1437  -1.262
## scale(education_years)                                0.9281     0.1566   5.926
## return_code1:poly(Age, 2)1                           40.6262    20.1991   2.011
## return_code1:poly(Age, 2)2                           34.7553    21.2138   1.638
```

```
## return_code1:scale(education_years)              -0.5794     0.9197  -0.630
## poly(Age, 2)1:scale(education_years)             -6.3192     4.2803  -1.476
## poly(Age, 2)2:scale(education_years)             -1.1259     4.0609  -0.277
## return_code1:poly(Age, 2)1:scale(education_years) -32.7095    28.4053  -1.152
## return_code1:poly(Age, 2)2:scale(education_years) -34.4891    29.4549  -1.171
##                                                  Pr(>|t|)
## (Intercept)                                       < 2e-16 ***
## return_code1                                       0.1169
## poly(Age, 2)1                                     < 2e-16 ***
## poly(Age, 2)2                                      0.2073
## scale(education_years)                           4.76e-09 ***
## return_code1:poly(Age, 2)1                         0.0447 *
## return_code1:poly(Age, 2)2                         0.1018
## return_code1:scale(education_years)                0.5289
## poly(Age, 2)1:scale(education_years)               0.1403
## poly(Age, 2)2:scale(education_years)               0.7817
## return_code1:poly(Age, 2)1:scale(education_years)   0.2499
## return_code1:poly(Age, 2)2:scale(education_years)   0.2420
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.716 on 742 degrees of freedom
## Multiple R-squared:  0.1807, Adjusted R-squared:  0.1685
## F-statistic: 14.88 on 11 and 742 DF,  p-value: < 2.2e-16
```

```
car::Anova(m_mem, type=3, statistic = F)
```

```
## Anova Table (Type III tests)
##
## Response: mem_imm
##                                                Sum Sq  Df   F value    Pr(>F)
## (Intercept)                                    117321   1 8497.9053 < 2.2e-16
## return_code                                        34   1    2.4637   0.11693
## poly(Age, 2)                                     1033   2   37.4178 3.295e-16
## scale(education_years)                            485   1   35.1143 4.759e-09
## return_code:poly(Age, 2)                           94   2    3.4221   0.03316
## return_code:scale(education_years)                  5   1    0.3969   0.52889
## poly(Age, 2):scale(education_years)                34   2    1.2386   0.29039
## return_code:poly(Age, 2):scale(education_years)    33   2    1.1835   0.30680
## Residuals                                       10244 742
##
## (Intercept)                                    ***
## return_code
## poly(Age, 2)                                   ***
## scale(education_years)                         ***
## return_code:poly(Age, 2)                       *
## return_code:scale(education_years)
## poly(Age, 2):scale(education_years)
## return_code:poly(Age, 2):scale(education_years)
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# selectivity age-orthogonal (and education-adjusted) for Del
clean_data <- na.omit(mydata1_R_NR[, c("mem_del", "Age", "education_years", "return_code")])
del_model <- lm(mem_del ~ poly(Age,2), data = clean_data)
clean_data$regressed_del <- residuals(del_model, na.rm = TRUE)

returners_residuals <- clean_data$regressed_del[clean_data$return_code == "1"]
cc700_residuals     <- clean_data$regressed_del
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)
```

```
## [1] 0.1914703
```

```r
m_mem <-lm(mem_del ~ return_code*poly(Age,2)*scale(education_years) , data = clean_data)
summary(m_mem)
```

```
##
## Call:
## lm(formula = mem_del ~ return_code * poly(Age, 2) * scale(education_years),
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -13.7106  -2.3883   0.2294   2.5909  12.0307
##
## Coefficients:
##                                                   Estimate Std. Error t value
## (Intercept)                                        12.5571     0.1603  78.339
## return_code1                                        0.9964     0.6971   1.429
## poly(Age, 2)1                                     -41.7200     4.2406  -9.838
## poly(Age, 2)2                                      -9.1790     4.3126  -2.128
## scale(education_years)                              0.9167     0.1630   5.624
## return_code1:poly(Age, 2)1                         46.0306    21.0225   2.190
## return_code1:poly(Age, 2)2                         46.4661    22.0786   2.105
## return_code1:scale(education_years)                -0.2610     0.9572  -0.273
## poly(Age, 2)1:scale(education_years)               -5.8245     4.4548  -1.307
## poly(Age, 2)2:scale(education_years)               -4.3988     4.2264  -1.041
## return_code1:poly(Age, 2)1:scale(education_years) -31.3588    29.5632  -1.061
## return_code1:poly(Age, 2)2:scale(education_years) -44.9730    30.6556  -1.467
##                                                   Pr(>|t|)
## (Intercept)                                        < 2e-16 ***
## return_code1                                        0.1533
## poly(Age, 2)1                                      < 2e-16 ***
## poly(Age, 2)2                                       0.0336 *
## scale(education_years)                            2.65e-08 ***
## return_code1:poly(Age, 2)1                          0.0289 *
## return_code1:poly(Age, 2)2                          0.0357 *
## return_code1:scale(education_years)                 0.7852
## poly(Age, 2)1:scale(education_years)                0.1915
## poly(Age, 2)2:scale(education_years)                0.2983
## return_code1:poly(Age, 2)1:scale(education_years)   0.2892
## return_code1:poly(Age, 2)2:scale(education_years)   0.1428
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 3.867 on 742 degrees of freedom
## Multiple R-squared:  0.2087, Adjusted R-squared:  0.1969
## F-statistic: 17.79 on 11 and 742 DF,  p-value: < 2.2e-16
```

```
car::Anova(m_mem, type=3, statistic = F)
```

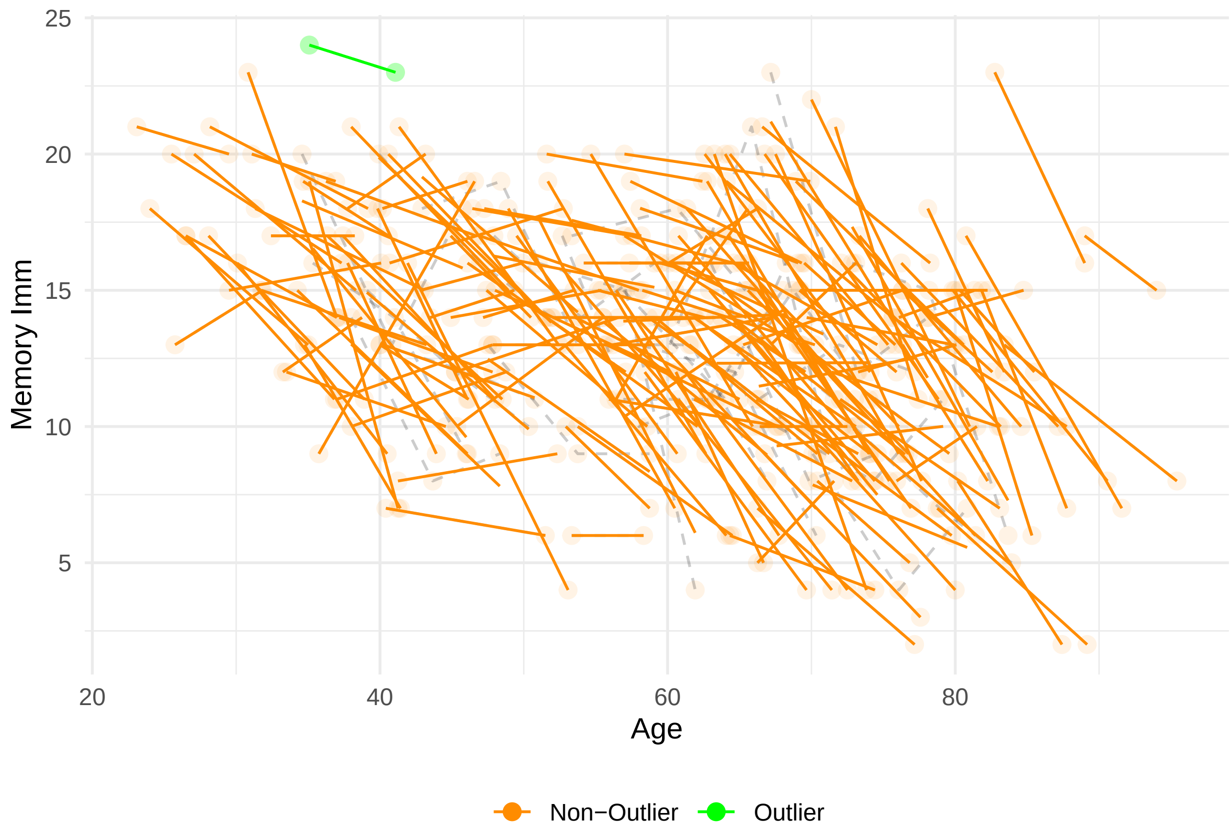
```
## Anova Table (Type III tests)
##
## Response: mem_del
##                                              Sum Sq  Df   F value    Pr(>F)
## (Intercept)                                   91774   1 6136.9587 < 2.2e-16
## return_code                                      31   1    2.0434  0.153287
## poly(Age, 2)                                   1514   2   50.6260 < 2.2e-16
## scale(education_years)                          473   1   31.6269 2.647e-08
## return_code:poly(Age, 2)                        140   2    4.6921  0.009441
## return_code:scale(education_years)                1   1    0.0743  0.785212
## poly(Age, 2):scale(education_years)              50   2    1.6880  0.185598
## return_code:poly(Age, 2):scale(education_years)  43   2    1.4498  0.235280
## Residuals                                     11096 742
##
## (Intercept)                                  ***
## return_code
## poly(Age, 2)                                 ***
## scale(education_years)                       ***
## return_code:poly(Age, 2)                     **
## return_code:scale(education_years)
## poly(Age, 2):scale(education_years)
## return_code:poly(Age, 2):scale(education_years)
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Memory ICC

```
df_mem <- mydata %>%
  filter(group %in% c("G2", "G3"),
         phase %in% c(1, 4, 5)) %>%
  group_by(CC.ID) %>%
  filter(sum(!is.na(mem_imm) & !is.na(mem_del)) >= 2) %>%
  ungroup()

df_mem <- df_mem %>%
  group_by(CC.ID) %>%
  mutate(A0 = as.numeric(Age[wave == '0']),
         dA = as.numeric(Age - A0))

df_mem$prac <- 1
df_mem$prac[str_detect(df_mem$wave, "0")] <- 0
df_mem$prac   <- factor(df_mem$prac)
df_mem$format <- factor(df_mem$format)
```

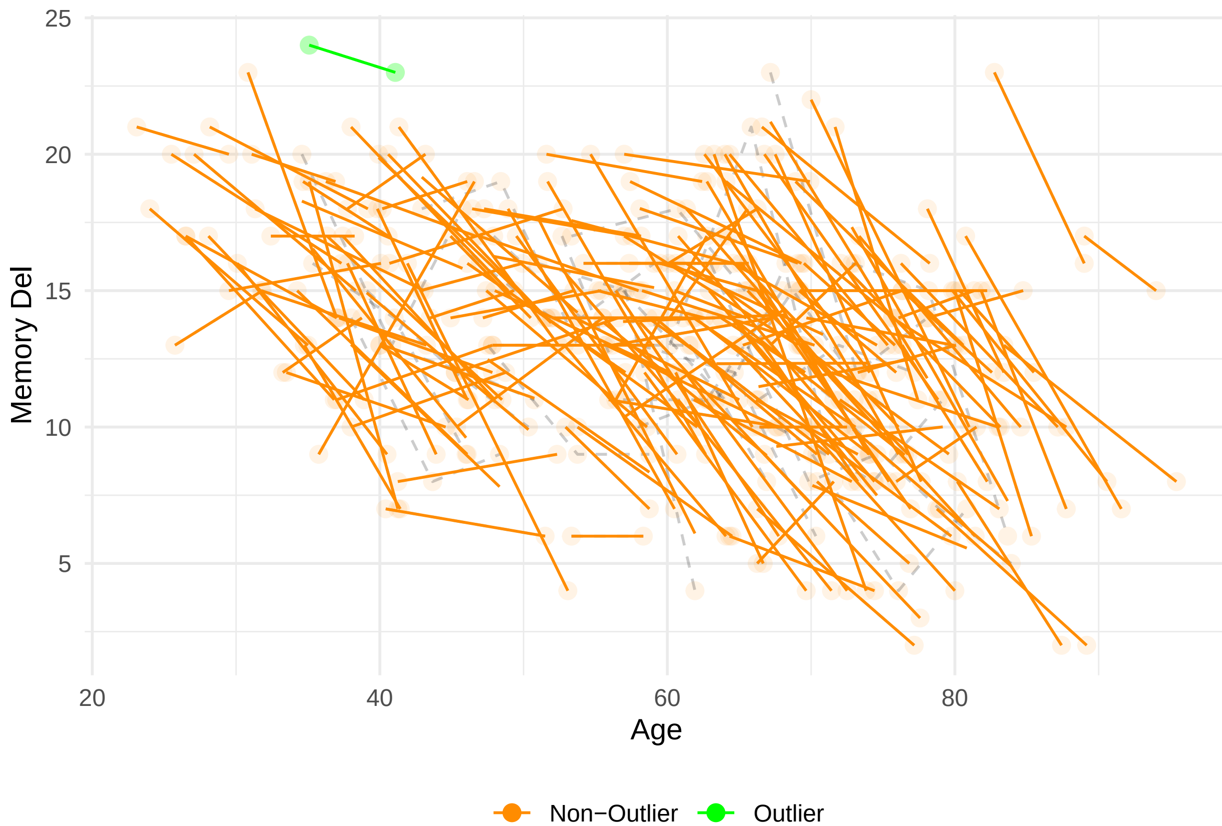
```r
df_mem$format <- ifelse(df_mem$wave == 0 & is.na(df_mem$format), 1, df_mem$format)

CC.ID <- df_mem$CC.ID[df_mem$wave == 0]
C0    <- df_mem$mem_imm[df_mem$wave == 0]
C1    <- df_mem$mem_imm[df_mem$wave == 1]
C2    <- df_mem$mem_imm[df_mem$wave == 2]
C0_1  <- df_mem$mem_del[df_mem$wave == 0]
C1_1  <- df_mem$mem_del[df_mem$wave == 1]
C2_1  <- df_mem$mem_del[df_mem$wave == 2]
educ  <- df_mem$education_years[df_mem$wave == 0]
sex   <- df_mem$sex[df_mem$wave == 0]
A0    <- df_mem$Age[df_mem$wave == 0]

feature_matrix <- data.frame(CC.ID, C0, C1, C2, C0_1, C1_1, C2_1, A0, sex, educ)

set.seed(987)
iso <- isolation.forest(feature_matrix[, c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0")], ntrees = 1(

feature_matrix$pred <- predict(iso, feature_matrix[, c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0")]
outliers <- feature_matrix[feature_matrix$pred > 0.60, c("CC.ID", "C0", "C1", "C2", "C0_1", "C1_1", "C2
outliers_mem <- outliers

plot_outliers(df_mem, "Age", "mem_imm", "Age", "Memory Imm", min(df_mem$mem_imm), max(df_mem$mem_imm), 
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 157 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 88 rows containing missing values or values outside the scale range
## (`geom_line()`).
```




```
plot_outliers(df_mem, "Age", "mem_imm", "Age", "Memory Del", min(df_mem$mem_del), max(df_mem$mem_del), c
```

```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 157 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 157 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 88 rows containing missing values or values outside the scale range
## (`geom_line()`).
```




```r
df_mem$mem_imm[df_mem$CC.ID %in% outliers_mem$CC.ID] <- NA
df_mem$mem_del[df_mem$CC.ID %in% outliers_mem$CC.ID] <- NA

m_imm = lmer('scale(mem_imm) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + forma
summary(m_imm)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(mem_imm) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + prac +
##    Data: df_mem
##
## REML criterion at convergence: 982.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.47473 -0.57057  0.04635  0.53628  2.45701
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  CC.ID    (Intercept) 0.2988   0.5467
##  Residual             0.4095   0.6399
## Number of obs: 399, groups:  CC.ID, 185
##
## Fixed effects:
```

```
##                                        Estimate Std. Error        df t value
## (Intercept)                            -0.14056    0.30975 337.32078  -0.454
## scale(A0)                              -0.25989    0.29667 338.24662  -0.876
## scale(dA)                              -0.23610    0.22842 261.34269  -1.034
## groupG3                                 0.02204    0.15765 346.43592   0.140
## scale(education_years)                  0.23952    0.08138 176.56351   2.943
## sex2                                    0.19388    0.10946 173.08829   1.771
## format                                  0.13283    0.19693 314.75061   0.675
## prac1                                  -0.18478    0.28486 274.34054  -0.649
## scale(A0):scale(dA)                    -0.01123    0.25256 276.02603  -0.044
## scale(A0):groupG3                       0.05186    0.16271 365.89241   0.319
## scale(dA):groupG3                      -0.11030    0.13748 235.63081  -0.802
## scale(A0):scale(education_years)       -0.03930    0.08962 179.07796  -0.439
## scale(A0):sex2                          0.04040    0.11094 179.13445   0.364
## scale(education_years):sex2            -0.05543    0.12279 177.56663  -0.451
## scale(A0):prac1                        -0.04377    0.31534 282.91966  -0.139
## scale(A0):scale(dA):groupG3            -0.07352    0.14004 248.30144  -0.525
## scale(A0):scale(education_years):sex2   0.07438    0.12215 179.77794   0.609
##                                       Pr(>|t|)
## (Intercept)                            0.65027
## scale(A0)                              0.38164
## scale(dA)                              0.30229
## groupG3                                0.88891
## scale(education_years)                 0.00368 **
## sex2                                   0.07828 .
## format                                 0.50048
## prac1                                  0.51709
## scale(A0):scale(dA)                    0.96457
## scale(A0):groupG3                      0.75009
## scale(dA):groupG3                      0.42318
## scale(A0):scale(education_years)       0.66151
## scale(A0):sex2                         0.71616
## scale(education_years):sex2            0.65224
## scale(A0):prac1                        0.88971
## scale(A0):scale(dA):groupG3            0.60006
## scale(A0):scale(education_years):sex2  0.54331
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```r
wide <- pivot_wider(df_mem[df_mem$group == "G3" & df_mem$wave %in% c(0, 2),],
  id_cols = CC.ID,
  names_from = wave,
  values_from = mem_imm)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
##  Single Score Intraclass Correlation
##
```

```
##    Model: twoway
##    Type : consistency
##
##    Subjects = 130
##      Raters = 2
##    ICC(C,1) = 0.472
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(129,129) = 2.79 , p = 6.09e-09
##
##  95%-Confidence Interval for ICC Population Values:
##   0.327 < ICC < 0.596
```

```r
wide <- pivot_wider(df_mem[df_mem$group == "G3" & df_mem$wave %in% c(0, 2),],
  id_cols = CC.ID,
  names_from = wave,
  values_from = mem_del)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
##  Single Score Intraclass Correlation
##
##    Model: twoway
##    Type : consistency
##
##    Subjects = 130
##      Raters = 2
##    ICC(C,1) = 0.564
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(129,129) = 3.59 , p = 1.12e-12
##
##  95%-Confidence Interval for ICC Population Values:
##   0.435 < ICC < 0.671
```

```r
# #sensitivity check
# df_sub <- df_mem[df_mem$group == "G3", ]
# mod <- lmer(mem_imm ~ 1 + (1 | CC.ID), data = df_sub)
# summary(mod)
# vc <- as.data.frame(VarCorr(mod))
# ICC_C <- vc$vcov[vc$grp == "CC.ID"] /
#          (vc$vcov[vc$grp == "CC.ID"] + vc$vcov[vc$grp == "Residual"])
# ICC_C
#
# #sensitivity check
# mod <- lmer(mem_imm ~ 1 + (1 | CC.ID), data = df_mem)
# summary(mod)
# vc <- as.data.frame(VarCorr(mod))
# var_between <- vc$vcov[vc$grp == "CC.ID"]
# var_within  <- vc$vcov[vc$grp == "Residual"]
# ICC_C <- var_between / (var_between + var_within)
# ICC_C
```

SPEED (inverse Reaction Time)

```
# outliers RT
outliers_rt_simple_R <- boxplot.stats(mydata1_R$simple_speed)$out
outliers_rt_simple_NR <- boxplot.stats(mydata1_NR$simple_speed)$out

mydata1_R_NR$simple_speed[mydata1_R_NR$return_code == 1 & mydata1_R_NR$simple_speed %in% outliers_rt_si
mydata1_R_NR$simple_speed[mydata1_R_NR$return_code == 0 & mydata1_R_NR$simple_speed %in% outliers_rt_si

outliers_rt_choice_R <- boxplot.stats(mydata1_R$choice_speed)$out
outliers_rt_choice_NR <- boxplot.stats(mydata1_NR$choice_speed)$out

mydata1_R_NR$choice_speed[mydata1_R_NR$return_code == 1 & mydata1_R_NR$choice_speed %in% outliers_rt_ch
mydata1_R_NR$choice_speed[mydata1_R_NR$return_code == 0 & mydata1_R_NR$choice_speed %in% outliers_rt_ch

wilcox.test(simple_speed ~ return_code, data = mydata1_R_NR)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  simple_speed by return_code
## W = 34850, p-value = 0.01523
## alternative hypothesis: true location shift is not equal to 0
```

```
wilcox.test(choice_speed ~ return_code, data = mydata1_R_NR)
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  choice_speed by return_code
## W = 35341, p-value = 0.02333
## alternative hypothesis: true location shift is not equal to 0
```

Speed Selectivity

```
# selectivity age-driven
(mean(mydata1_R$simple_speed, na.rm = TRUE) - (mean(mydata1_R_NR$simple_speed, na.rm = TRUE)))/sd(mydata
```

```
## [1] 0.1648978
```

```
(mean(mydata1_R$choice_speed, na.rm = TRUE) - (mean(mydata1_R_NR$choice_speed, na.rm = TRUE)))/sd(mydata
```

```
## [1] 0.1679166
```

```
# selectivity age-orthogonal
clean_data <- na.omit(mydata1_R_NR[, c("simple_speed", "education_years", "return_code", "Age", "sex")]

rt_model <- lm(simple_speed ~ poly(Age,2), data = clean_data)
```

```r
clean_data$regressed_simple <- residuals(rt_model, na.rm = TRUE)
returners_residuals <- clean_data$regressed_simple[clean_data$return_code == "1"]
cc700_residuals <- clean_data$regressed_simple
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)
```

```
## [1] 0.1784335
```

```r
rt_m <-lm(simple_speed ~ return_code*poly(Age,2)*scale(education_years) , data = clean_data)
summary(rt_m)
```

```
##
## Call:
## lm(formula = simple_speed ~ return_code * poly(Age, 2) * scale(education_years),
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.23172 -0.27279  0.04253  0.30634  1.16184
##
## Coefficients:
##                                                 Estimate Std. Error t value
## (Intercept)                                      2.80118    0.01821 153.786
## return_code1                                     0.01800    0.07865   0.229
## poly(Age, 2)1                                   -4.01975    0.46436  -8.657
## poly(Age, 2)2                                   -0.68123    0.47288  -1.441
## scale(education_years)                           0.08284    0.01857   4.462
## return_code1:poly(Age, 2)1                       1.58479    2.30148   0.689
## return_code1:poly(Age, 2)2                       0.31077    2.42710   0.128
## return_code1:scale(education_years)              0.05620    0.10764   0.522
## poly(Age, 2)1:scale(education_years)            -0.88565    0.48564  -1.824
## poly(Age, 2)2:scale(education_years)            -0.55513    0.45980  -1.207
## return_code1:poly(Age, 2)1:scale(education_years) -1.41764    3.23654  -0.438
## return_code1:poly(Age, 2)2:scale(education_years) -0.36913    3.36720  -0.110
##                                                 Pr(>|t|)
## (Intercept)                                      < 2e-16 ***
## return_code1                                      0.8190
## poly(Age, 2)1                                    < 2e-16 ***
## poly(Age, 2)2                                     0.1502
## scale(education_years)                          9.49e-06 ***
## return_code1:poly(Age, 2)1                        0.4913
## return_code1:poly(Age, 2)2                        0.8982
## return_code1:scale(education_years)               0.6017
## poly(Age, 2)1:scale(education_years)              0.0686 .
## poly(Age, 2)2:scale(education_years)              0.2277
## return_code1:poly(Age, 2)1:scale(education_years)   0.6615
## return_code1:poly(Age, 2)2:scale(education_years)   0.9127
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.422 on 687 degrees of freedom
## Multiple R-squared:  0.1725, Adjusted R-squared:  0.1592
## F-statistic: 13.02 on 11 and 687 DF,  p-value: < 2.2e-16
```

```r
car::Anova(rt_m, type=3, statistic = F)
```

```
## Anova Table (Type III tests)
##
## Response: simple_speed
##                                                Sum Sq  Df    F value    Pr(>F)
## (Intercept)                                    4211.0   1 23650.2553 < 2.2e-16
## return_code                                       0.0   1     0.0524   0.81900
## poly(Age, 2)                                     13.7   2    38.3732 < 2.2e-16
## scale(education_years)                            3.5   1    19.9086  9.49e-06
## return_code:poly(Age, 2)                          0.1   2     0.2513   0.77786
## return_code:scale(education_years)                0.0   1     0.2726   0.60174
## poly(Age, 2):scale(education_years)               1.0   2     2.8872   0.05641
## return_code:poly(Age, 2):scale(education_years)   0.0   2     0.0974   0.90719
## Residuals                                       122.3 687
##
## (Intercept)                                    ***
## return_code
## poly(Age, 2)                                   ***
## scale(education_years)                         ***
## return_code:poly(Age, 2)
## return_code:scale(education_years)
## poly(Age, 2):scale(education_years)            .
## return_code:poly(Age, 2):scale(education_years)
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# selectivity age-orthogonal choice
clean_data <- na.omit(mydata1_R_NR[, c("choice_speed", "education_years", "return_code", "Age")])

rt_model <- lm(choice_speed ~ poly(Age,2), data = clean_data)
clean_data$regressed_choice <- residuals(rt_model, na.rm = TRUE)
returners_residuals <- clean_data$regressed_choice[clean_data$return_code == "1"]
cc700_residuals <- clean_data$regressed_choice
(mean(returners_residuals) - mean(cc700_residuals)) / sd(cc700_residuals)
```

```
## [1] 0.197251
```

```r
rt_m <-lm(choice_speed ~ return_code*poly(Age,2)*scale(education_years) , data = clean_data)
summary(rt_m)
```

```
##
## Call:
## lm(formula = choice_speed ~ return_code * poly(Age, 2) * scale(education_years),
##     data = clean_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.94910 -0.18524  0.01429  0.19418  0.78503
##
## Coefficients:
```

```
##                                                  Estimate Std. Error t value
## (Intercept)                                       1.80960    0.01191 151.977
## return_code1                                      0.07565    0.05105   1.482
## poly(Age, 2)1                                    -5.92328    0.30331 -19.529
## poly(Age, 2)2                                     0.42911    0.30914   1.388
## scale(education_years)                            0.06736    0.01211   5.562
## return_code1:poly(Age, 2)1                       -1.96860    1.49116  -1.320
## return_code1:poly(Age, 2)2                        1.40502    1.56849   0.896
## return_code1:scale(education_years)              -0.08302    0.06923  -1.199
## poly(Age, 2)1:scale(education_years)             -0.85998    0.31638  -2.718
## poly(Age, 2)2:scale(education_years)             -0.05743    0.30157  -0.190
## return_code1:poly(Age, 2)1:scale(education_years)  1.19969    2.06502   0.581
## return_code1:poly(Age, 2)2:scale(education_years) -2.75877    2.17007  -1.271
##                                                  Pr(>|t|)
## (Intercept)                                       < 2e-16 ***
## return_code1                                      0.13884
## poly(Age, 2)1                                     < 2e-16 ***
## poly(Age, 2)2                                     0.16556
## scale(education_years)                           3.82e-08 ***
## return_code1:poly(Age, 2)1                        0.18722
## return_code1:poly(Age, 2)2                        0.37068
## return_code1:scale(education_years)               0.23084
## poly(Age, 2)1:scale(education_years)              0.00673 **
## poly(Age, 2)2:scale(education_years)              0.84903
## return_code1:poly(Age, 2)1:scale(education_years)  0.56146
## return_code1:poly(Age, 2)2:scale(education_years)  0.20406
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.2753 on 687 degrees of freedom
## Multiple R-squared:  0.4571, Adjusted R-squared:  0.4484
## F-statistic: 52.59 on 11 and 687 DF,  p-value: < 2.2e-16
```

```
car::Anova(rt_m, type=3, statistic = F)
```

```
## Anova Table (Type III tests)
##
## Response: choice_speed
##                                                Sum Sq  Df    F value
## (Intercept)                                   1750.29   1 23096.9500
## return_code                                      0.17   1     2.1959
## poly(Age, 2)                                    29.09   2   191.9520
## scale(education_years)                           2.34   1    30.9351
## return_code:poly(Age, 2)                         0.18   2     1.2103
## return_code:scale(education_years)               0.11   1     1.4382
## poly(Age, 2):scale(education_years)              0.59   2     3.9123
## return_code:poly(Age, 2):scale(education_years)  0.16   2     1.0744
## Residuals                                       52.06 687
##                                                   Pr(>F)
## (Intercept)                                   < 2.2e-16 ***
## return_code                                     0.13884
## poly(Age, 2)                                  < 2.2e-16 ***
## scale(education_years)                        3.824e-08 ***
## return_code:poly(Age, 2)                        0.29875
```

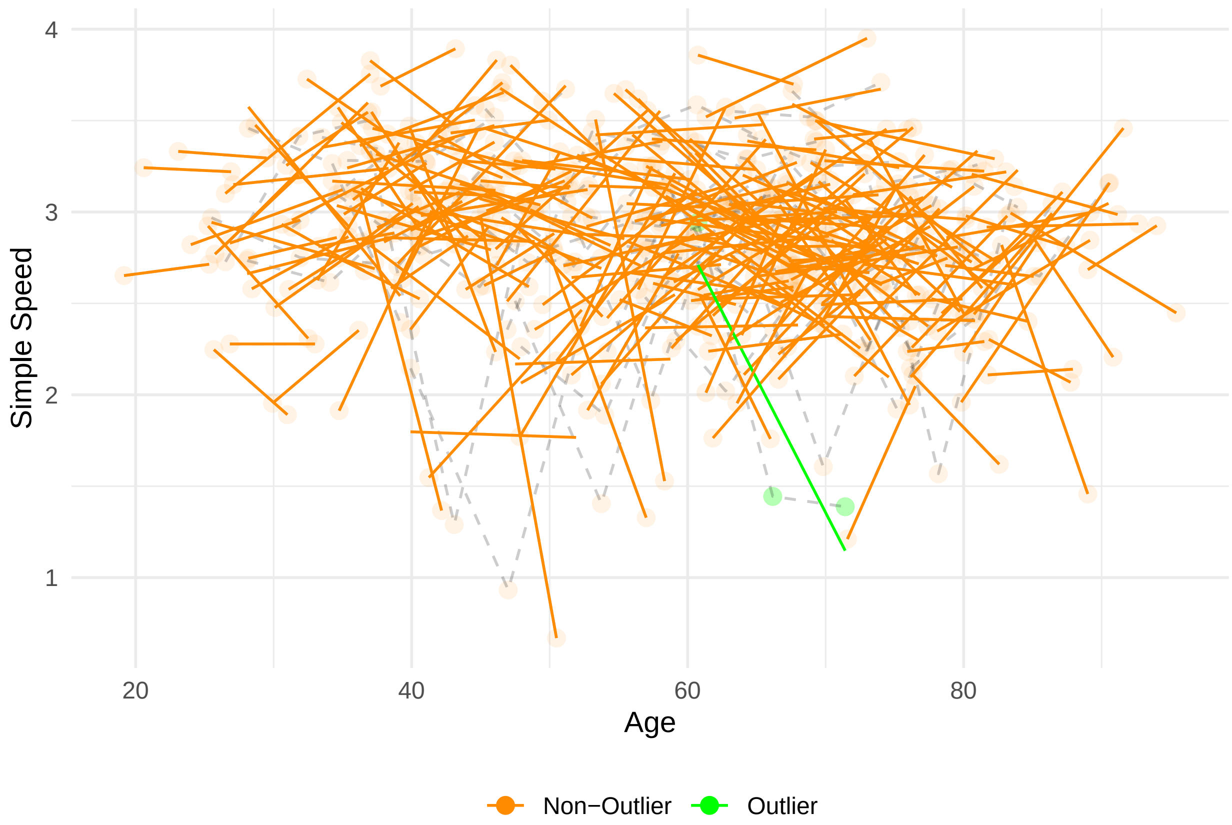
```
## return_code:scale(education_years)                0.23084
## poly(Age, 2):scale(education_years)               0.02044 *
## return_code:poly(Age, 2):scale(education_years)   0.34206
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Speed ICC

```
df_speed <- mydata %>%
  filter(group %in% c("G2", "G3"),
         phase %in% c(1, 4, 5)) %>%
  group_by(CC.ID) %>%
  filter(sum(!is.na(simple_speed) & !is.na(choice_speed)) >= 2) %>%
  ungroup()

df_speed <- df_speed %>%
  group_by(CC.ID) %>%
  mutate(A0 = as.numeric(Age[wave == '0']),
         dA = as.numeric(Age - A0))

df_speed$prac <- 1
df_speed$prac[str_detect(df_speed$wave, "0")] <- 0
df_speed$prac   <- factor(df_speed$prac)
df_speed$format <- factor(df_speed$format)
df_speed$format <- ifelse(df_speed$wave == 0 & is.na(df_speed$format), 1, df_speed$format)

CC.ID <- df_speed$CC.ID[df_speed$wave == 0]
C0    <- df_speed$simple_speed[df_speed$wave == 0]
C1    <- df_speed$simple_speed[df_speed$wave == 1]
C2    <- df_speed$simple_speed[df_speed$wave == 2]
C0_1  <- df_speed$choice_speed[df_speed$wave == 0]
C1_1  <- df_speed$choice_speed[df_speed$wave == 1]
C2_1  <- df_speed$choice_speed[df_speed$wave == 2]
educ  <- df_speed$education_years[df_speed$wave == 0]
sex   <- df_speed$sex[df_speed$wave == 0]
A0    <- df_speed$Age[df_speed$wave == 0]

feature_matrix <- data.frame(CC.ID, C0, C1, C2, C0_1, C1_1, C2_1, A0, sex, educ)

set.seed(987)
iso <- isolation.forest(feature_matrix[, c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0")], ntrees = 1

feature_matrix$pred <- predict(iso, feature_matrix[, c("C0", "C1", "C2", "C0_1", "C1_1", "C2_1", "A0")]
outliers <- feature_matrix[feature_matrix$pred > 0.60, c("CC.ID", "C0", "C1", "C2", "C0_1", "C1_1", "C2
outliers_speed <- outliers

plot_outliers(df_speed, "Age", "simple_speed", "Age", "Simple Speed", min(df_speed$simple_speed), max(d
```

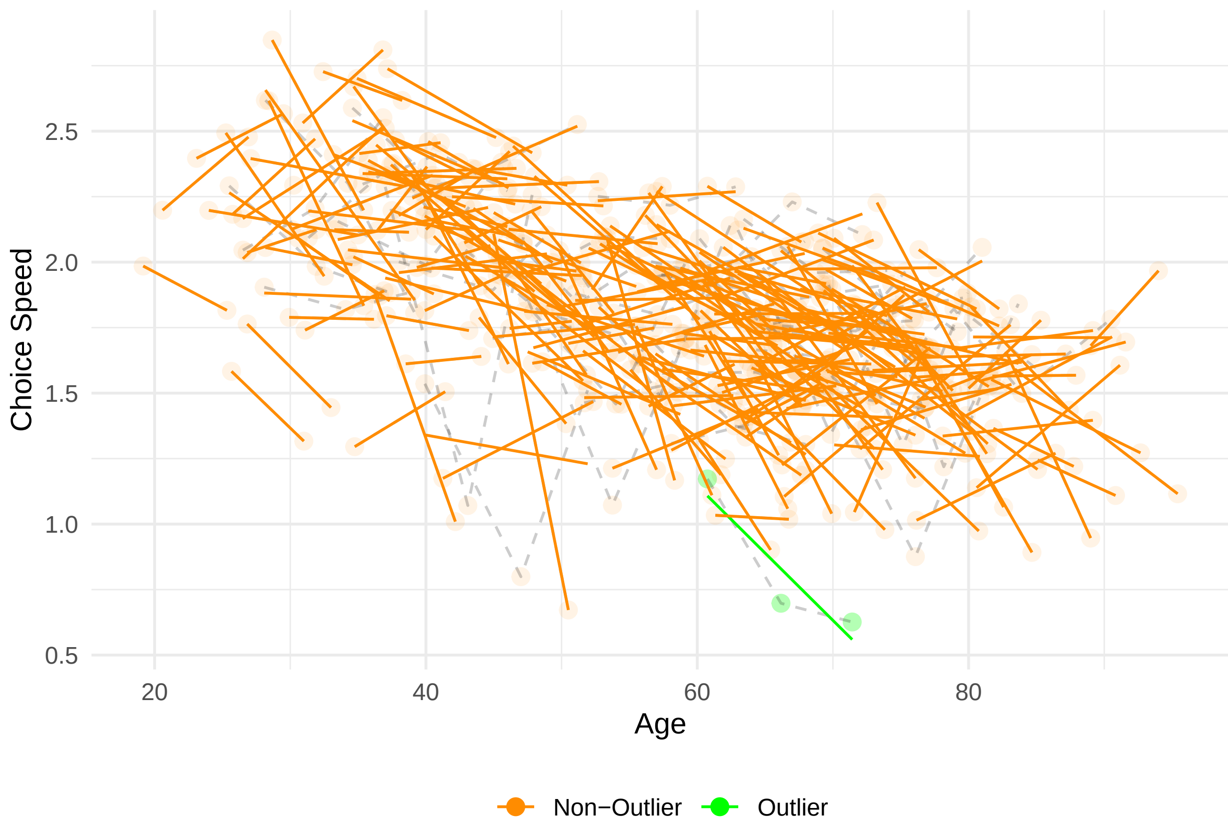
```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: Removed 207 rows containing non-finite outside the scale range
```

```
## ('stat_smooth()').

## Warning: Removed 207 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 163 rows containing missing values or values outside the scale range
## ('geom_line()').
```




```r
plot_outliers(df_speed, "Age", "choice_speed", "Age", "Choice Speed", min(df_speed$choice_speed), max(d
```

```
## 'geom_smooth()' using formula = 'y ~ x'

## Warning: Removed 210 rows containing non-finite outside the scale range
## ('stat_smooth()').

## Warning: Removed 210 rows containing missing values or values outside the scale range
## ('geom_point()').

## Warning: Removed 164 rows containing missing values or values outside the scale range
## ('geom_line()').
```




```r
df_speed$simple_speed[df_speed$CC.ID %in% outliers_speed$CC.ID] <- NA
df_speed$choice_speed[df_speed$CC.ID %in% outliers_speed$CC.ID] <- NA

m_simple = lmer('scale(simple_speed) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex
summary(m_simple)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(simple_speed) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + p
##    Data: df_speed
##
## REML criterion at convergence: 1546.1
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -4.3770 -0.4813  0.1005  0.5676  1.8831
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  CC.ID    (Intercept) 0.2240   0.4733
##  Residual             0.7129   0.8443
## Number of obs: 552, groups:  CC.ID, 253
##
## Fixed effects:
```

```
##                                        Estimate Std. Error         df t value
## (Intercept)                            5.287e-01  2.787e-01  4.945e+02   1.897
## scale(A0)                             -2.823e-01  2.411e-01  4.887e+02  -1.171
## scale(dA)                              5.247e-01  2.057e-01  4.047e+02   2.551
## groupG3                               -7.986e-02  1.243e-01  4.471e+02  -0.642
## scale(education_years)                 1.129e-01  7.858e-02  2.561e+02   1.436
## sex2                                  -2.284e-01  9.830e-02  2.509e+02  -2.324
## format                                 1.523e-01  1.871e-01  5.048e+02   0.814
## prac1                                 -9.277e-01  2.696e-01  4.212e+02  -3.441
## scale(A0):scale(dA)                   -4.830e-02  2.196e-01  4.105e+02  -0.220
## scale(A0):groupG3                      8.520e-02  1.289e-01  4.498e+02   0.661
## scale(dA):groupG3                     -7.908e-02  1.235e-01  3.625e+02  -0.640
## scale(A0):scale(education_years)      -6.539e-02  7.999e-02  2.587e+02  -0.817
## scale(A0):sex2                         6.072e-02  9.955e-02  2.524e+02   0.610
## scale(education_years):sex2            3.482e-02  1.067e-01  2.594e+02   0.326
## scale(A0):prac1                       -5.148e-04  2.878e-01  4.187e+02  -0.002
## scale(A0):scale(dA):groupG3            7.827e-02  1.255e-01  3.730e+02   0.624
## scale(A0):scale(education_years):sex2  9.739e-02  1.077e-01  2.619e+02   0.904
##                                       Pr(>|t|)
## (Intercept)                           0.058411 .
## scale(A0)                             0.242296
## scale(dA)                             0.011100 *
## groupG3                               0.520993
## scale(education_years)                0.152123
## sex2                                  0.020943 *
## format                                0.416072
## prac1                                 0.000636 ***
## scale(A0):scale(dA)                   0.825991
## scale(A0):groupG3                     0.508899
## scale(dA):groupG3                     0.522271
## scale(A0):scale(education_years)      0.414404
## scale(A0):sex2                        0.542490
## scale(education_years):sex2           0.744482
## scale(A0):prac1                       0.998574
## scale(A0):scale(dA):groupG3           0.533336
## scale(A0):scale(education_years):sex2 0.366657
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

##
## Correlation matrix not shown by default, as p = 17 > 12.
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```r
m_choice = lmer('scale(choice_speed) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex
summary(m_choice)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## "scale(choice_speed) ~ scale(A0)*scale(dA)*group + scale(A0)*scale(education_years)*sex + format + p
##    Data: df_speed
##
```

```
## REML criterion at convergence: 1270.5
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -3.8069 -0.4934  0.0546  0.5623  1.9303
##
## Random effects:
##  Groups   Name        Variance Std.Dev.
##  CC.ID    (Intercept) 0.2798   0.5290
##  Residual             0.3447   0.5871
## Number of obs: 549, groups:  CC.ID, 253
##
## Fixed effects:
##                                        Estimate Std. Error        df t value
## (Intercept)                             0.20336    0.20881 471.07258   0.974
## scale(A0)                              -0.70391    0.18317 478.63889  -3.843
## scale(dA)                               0.09072    0.15009 359.36435   0.604
## groupG3                                 0.03592    0.10349 411.59523   0.347
## scale(education_years)                  0.11852    0.06984 247.38665   1.697
## sex2                                   -0.08867    0.08746 244.25091  -1.014
## format                                  0.18613    0.13962 442.02847   1.333
## prac1                                  -0.67185    0.19847 370.47235  -3.385
## scale(A0):scale(dA)                    -0.07713    0.16141 362.77159  -0.478
## scale(A0):groupG3                       0.01455    0.10744 420.07799   0.135
## scale(dA):groupG3                       0.04453    0.08835 332.94876   0.504
## scale(A0):scale(education_years)       -0.03590    0.07101 247.90000  -0.506
## scale(A0):sex2                          0.17453    0.08855 244.88345   1.971
## scale(education_years):sex2             0.05000    0.09468 248.75927   0.528
## scale(A0):prac1                         0.05741    0.21270 367.54567   0.270
## scale(A0):scale(dA):groupG3             0.07926    0.09056 340.41478   0.875
## scale(A0):scale(education_years):sex2   0.08785    0.09547 249.42848   0.920
##                                       Pr(>|t|)
## (Intercept)                           0.330594
## scale(A0)                             0.000138 ***
## scale(dA)                             0.545918
## groupG3                               0.728688
## scale(education_years)                0.090943 .
## sex2                                  0.311678
## format                                0.183175
## prac1                                 0.000787 ***
## scale(A0):scale(dA)                   0.633031
## scale(A0):groupG3                     0.892331
## scale(dA):groupG3                     0.614582
## scale(A0):scale(education_years)      0.613591
## scale(A0):sex2                        0.049849 *
## scale(education_years):sex2           0.597870
## scale(A0):prac1                       0.787386
## scale(A0):scale(dA):groupG3           0.382082
## scale(A0):scale(education_years):sex2 0.358347
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Correlation matrix not shown by default, as p = 17 > 12.
```

```
## Use print(x, correlation=TRUE)  or
##     vcov(x)        if you need it
```

```r
wide <- pivot_wider(df_speed[df_speed$group == "G3" & df_speed$wave %in% c(0, 2),],
  id_cols = CC.ID,
  names_from = wave,
  values_from = simple_speed)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
##  Single Score Intraclass Correlation
##
##    Model: twoway
##    Type : consistency
##
##    Subjects = 133
##      Raters = 2
##    ICC(C,1) = 0.155
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(132,132) = 1.37 , p = 0.0373
##
##  95%-Confidence Interval for ICC Population Values:
##   -0.015 < ICC < 0.316
```

```r
wide <- pivot_wider(df_speed[df_speed$group == "G3" & df_speed$wave %in% c(0, 2),],
  id_cols = CC.ID,
  names_from = wave,
  values_from = choice_speed)
mat <- wide[, -1]
icc(mat, model = "twoway", type = "consistency", unit = "single")
```

```
##  Single Score Intraclass Correlation
##
##    Model: twoway
##    Type : consistency
##
##    Subjects = 132
##      Raters = 2
##    ICC(C,1) = 0.663
##
##  F-Test, H0: r0 = 0 ; H1: r0 > 0
##  F(131,131) = 4.93 , p = 1.83e-18
##
##  95%-Confidence Interval for ICC Population Values:
##   0.555 < ICC < 0.749
```